

Every Drop is Precious

JAL DIGDARSHIKA

(GUIDE TO WATER)

Author: Raman Kant
Riverman of India



JAL AYAM
PARYAVARAN SAMRAKSHAN GATIVIDHI
MEERUT REGION

EVERY DROP IS PRECIOUS....

JAL DIGDARSHIKA

(GUIDE TO WATER)



JAL AYAM
PARYAVARAN SAMRAKSHAN GATIVIDHI
MEERUT REGION

**JAL AYAM
PARYAVARAN SAMRAKSHAN GATIVIDHI
MEERUT REGION**

Author/compiler/illustrator

RAMAN KANT

Riverman of India

Special Support

Dr. Sanjay Gupta

Mr. Mukesh Tyagi

@ पर्यावरण संरक्षण गतिविधि



web: www.paryavaransamrakshan.org
E-mail: paryavaransamrakshan@gmail.com



JAL DIGDARSHIKA (GUIDE TO WATER)

Author

Raman Kant
Riverman of India





RAMAN KANT

Riverman of India

While 18 percent of the total world's population resides in India, it has only 2.4 percent of the land area of the world. This is one of the main reasons that India is a densely populated country, and there is a lot of pressure on water resources. This pressure is also high because India has only about 4 percent of the total water in the world. Out of the total groundwater extracted daily in the whole world, India's contribution is about 25 percent!

India has witnessed remarkable economic growth in the last three decades. Rapidly growing economics, population growth, and urbanization have had a great impact on the demand for water. Changes in food consumption, lifestyle, and land use patterns have also played a major role in increasing the demand for water. Although India receives plenty of water in the form of rainfall during the monsoon, only a small percentage of that water is added to the reserve due to a lack of storage. The spatial distribution of the amount of water available is highly mismatched with the population. There is also an irony that where more people live, less water is available.

At present, the availability of water in India has reduced to one-third compared to the year 1952, while the population has increased from 36 crores to about 135 crores. The situation has become such that we are continuously becoming more dependent on underground water resources. This has led to the underground water level slipping down at an average rate of one foot per year. Due to this, about 15 crore people in North India are facing a severe water crisis. Continuous exploitation of groundwater sources has resulted in declaring one-fourth of the developmental blocks of the country under dark zone, and the numbers are rapidly increasing. No part of the country has been spared from this situation. From north to south and from east to west, there is a struggle over water everywhere. From villages to cities and from small to big cities, the practice of tankers and tankers for drinking water has become common. In some parts of the country, standing in queues for water has become a daily routine for people. The situation has become even worse in villages. Earlier, water was available at an average distance of five minutes. Today, one has to walk at least half an hour for that. This condition prevails in abundance in the villages of Rajasthan, Gujarat, and some parts of the states of South India. Under the 'Har Ghar Ko Nal Se Jal' scheme of the Government of India, efforts are being made to make clean drinking water easily available to every citizen of the country. According to NASA's Satellite Gravity Recovery and Climate Experiment, groundwater reserves in Punjab, Haryana, and Rajasthan are depleting very fast. The World Bank estimates that 60 percent of the world's groundwater resources will be at risk in the next 25 years. The situation is all the more worrying for India because about 70 percent of our water demand is met from groundwater sources. If this happens, there will be no water to grow crops or to make goods in factories. Agriculture and industries will not only be ruined, but a large part of our population will also yearn for every drop of water.

According to the study of the National Commission for Integrated Water Resources Development Plan and the Center for Science and Environment, the total groundwater demand in the industrial and energy sector will increase to 8.5 percent by the year 2025 as against 7 percent today, which will increase to about 10.1 percent by the year 2050. Through this, it can be easily understood



how important it is to save groundwater for the future. According to a study, the demand for groundwater in the agriculture sector will decrease to 70.9 percent by the year 2050 compared to 77.3 percent in the year 2010, that is, in the future where the demand for water will increase in the industry and energy sector, it will decrease in agriculture.

Delhi and its nearby states have experienced mild and short-duration tremors about two dozen times in the last one decade. This is a clear sign of groundwater exploitation as the water below the ground is constantly drying up. The water reservoirs are being drained, and due to the depletion of groundwater reserves, the tectonic plates beneath the earth are slowly moving. If the situation continues to remain the same, then the possibility of a big earthquake in the near future cannot be ruled out. There is an urgent need for stricter laws for groundwater use in India because except for some big cities, clean drinking water is used in the entire country for everything ranging from industry to agriculture and from drinking to flushing the toilet. In order to preserve groundwater, this habit needs to be changed. We should use purified and treated water from the sewage water plants for irrigation, industrial use, and also for day-to-day domestic activities (washing dishes, washing cars, in pots and toilets etc.) and along with this, factories and industries should be forced to reuse water. To save and conserve groundwater, every person will have to reduce their individual consumerism. We must be aware and realize how much water is being used to make everything that we use and whether our lives can go on without it or not. If people understand this and suitable strict laws are made and implemented for the exploitation of groundwater, then the future of water will be prosperous.

According to data for the year 2019, there are about 6,64,369 villages in India and 4000 small and big cities. According to an estimate, there are about 40 lakh reservoirs (different types of water sources) in the country, out of which about 30 percent have lost their entire existence, while about 90 percent of the remaining reservoirs are facing encroachment and only 10 percent have survived in their original form. At present, about 50 percent of the total reservoirs have dried up and 30 percent are ringing with dirt. 80 percent of the total existing reservoirs have lost their water storage capacity. The total length of rivers and canals in the country is about 1 lakh 8 thousand kilometers. There are 12 major river basins in India. A river basin is considered as the basic hydrological unit for planning and development of water resources. The total catchment area of these rivers is 25.3 lakh sq km. The largest river basin in India is the Ganga-Brahmaputra Meghna River basin, which covers about 11.0 lakh sq.km. Other major river basins are the Indus, Mahanadi, Godavari, and Krishna with a catchment area of more than 100,000 sq km. There are 46 medium river basins with a catchment area of between 2000 to 20000 sq km. The total catchment area of medium river basins is about 2.5 lakh sq.km. All the major river basins and several medium river basins flow inter-state and cover about 81 percent of the total geographical area of the country. Beneath our feet lies the treasure of groundwater which enriches our lives. Groundwater remains out of our sight but always persists in the mind. In other words, groundwater is the protector of our present and future.

Through this handbook, not only the experts in the field of water of each state in India will be helped to create a better understanding of water, its pollution, its conservation and river regeneration, but information will also be provided on the various methods of working at the ground level in each and every district of the country. By reading this booklet, any person can easily work on water conservation and river rejuvenation in his village or city. The purpose of publishing this booklet on various dimensions of water under environmental protection activity is to enable every worker to create his own understanding and do his job accordingly.

For writing this booklet, while inspiration and support was received from Mr. Ramavatar ji, the coordinator of environmental protection activities of Meerut, on the other hand, continuous guidance was received from Mr. Gopal Arya, the all-India coordinator of environmental protection activities and Mr. Rakesh Jain, the all-India co-coordinator. In preparing this booklet, the cooperation of many workers of environment protection activities of Meerut is also commendable.

This book is presented with the hope that various people associated with the water department of each state would make their respective districts richer in water by using this booklet along with their hard work and understanding. Although efforts have been made to provide the readers with a better insight, but still if there is any error, readers are most welcome to inform about it. With this, we will get an opportunity to rectify our mistakes and prevent them in future.





RAKESH JAIN

**All India Co-Convener
Environment Protection Initiative**

Five years ago, the Rashtriya Swayamsevak Sangh (RSS) launched an environment protection initiative at the national level to address the deepening water crisis, environmental degradation, and growing plastic waste in India. In this short period, the initiative has successfully reached out to environmental enthusiasts across the country. The aim of the RSS's environment protection initiative is to collaborate with organizations and individuals working to address the water, environment, and plastic waste crisis at the societal level in India. The Environment Protection Initiative is successfully carrying out this task.

In 2022, Jal Aayam, Meerut Region, initiated a positive effort to address the water crisis in India at the village, household, and individual levels. This initiative took the form of a "Jal Digidarshika" (Water Guide). This booklet presents water-related topics and methods for water conservation in a simple and easy-to-understand manner. The Hindi version of the booklet was published in June 2022. The booklet was prepared by Raman Kant, who was then the Jal Aayam Pramukh of Meerut Region. In the past two years, the booklet has been used to promote the work of Jal Aayam in other provinces of the country.

I am delighted to learn that the booklet is being translated into English and republished. This will make the booklet accessible to the English-speaking community and assist in promoting ground-level water conservation efforts.

I am confident that this booklet will achieve its goals.



Table of **CONTENTS**

Chapter - 1 WATER	1
Chapter - 2 RAINWATER HARVESTING	15
Chapter - 3 WATER POLLUTION	23
Chapter - 4 RIVER	33
Chapter - 5 HOW TO CONSERVE WATER	39

Water

CHAPTER ONE

Water is a liquid that provides us with physical energy. Pure water is colorless, odorless, and tasteless. In 1781, scientist Henry Vandy proposed that water is formed by the combination of hydrogen and oxygen. In 1804, French scientist Joseph Louis Gueux and German scientist Alexander von Humboldt confirmed through their studies that water is composed of two volumes of hydrogen and one volume of oxygen. Its chemical formula is H_2O (HO). Water is a transparent liquid. In 1783, senior French scientist Antoine Von stated that 'water is not an element but a compound'.

ORIGIN OF WATER ON EARTH

How and why water was created on Earth is not very clear. Indian sages have pondered deeply about the origin of water. They consider water as Narayan in its original form, born from Purushottam (Nara), hence called Nara. Before the creation, water (Naar) was the abode (Ayan) of God. Narayana means the abode of the Lord, as God resides in water. Water is considered indestructible, eternal, and infinite. The following Sanskrit verse describes water:

आपो नारा इत प्रोक्ता आपो वै नरसूनवः।
अयनं तस्य ताः पर्व तेन नारायणः स्मृतः ॥
- Manusmriti

This verse means that "You" (referring to various types of water) are called "Narah" because it



originated from "Nara." Since the original abode of "Nar" is in "Jal" (water), "Nara" who resides in water is called "Narayan."

The Indian philosophy considers water as a powerful and eternal substance. It is believed to have existed before the creation, remains present in the present time, and will continue to exist even after the destruction of the creation. According to ancient Indian hydrologists, water, known as 'Jal,' emerged from the mutual disturbance of the sky, air, and 'Tejas' (fire or electricity). In Indian philosophy, the sky is considered the father of the five great elements, and without it, sound would not be generated or heard. Vayu (air) is described as the only Parabrahma (supreme reality) before the creation, and its effulgence contributed to the origin of the world. 'Tejas' is synonymous with fire, electricity, or lightning. Water, therefore, originated from the mutual disturbance or explosion of the sky, air, and fire.

It is worth noting that understanding the ancient Indian philosophy regarding the birth of water is challenging, as it is presented in an adhidaivik (divine), adhibhautik (physical), and spiritual style.

Serious efforts are required to comprehend it fully. According to American scientist Mike Drake, water has been present since the birth of the earth. He suggests that water was already contained within the dust particles of the solar system during the formation of the earth. This hypothesis relies on the possibility of dust particles storing water in extreme conditions at the time of planet formation. Some scientists propose that after the birth of the earth, it was showered by billions of comets and meteorites saturated with water. Water accumulated on the earth's surface from these comets and meteorites, giving rise to the oceans. According to recent astrophysical discoveries, water reached earth from the outer edges of the solar system. It is believed that there was very little, if any, water on Earth at its birth. The arrival of moisture on earth is attributed to comets and water-laden meteorites located beyond Neptune. Scientists estimate that water arrived on earth approximately 400 million years ago.

American geologists have discovered a rock called Ringwoodite about 700 kilometers below the surface of the American continent, which contains vast reservoirs of water. They believe that water seeped through this rock and came to earth, as the pressure and temperature at a depth of 700 km are suitable for seepage. However, this discovery is based on circumstantial evidence and provides information specifically about the American continent, while conditions beneath other continents remain unknown.

The Big Bang event represents the concept of the early evolution of the universe.



Some scientists attempt to link the arrival of water on Earth with the Big Bang event, suggesting that the universe, initially created from energy, was extremely hot and heavy. It can be compared to a Black Hole.

About 1.37 billion years ago, the universe began expanding, resulting in a decrease in temperature and density. This expansion generated immense energy, causing a large area of space to increase in temperature. As a result, the space became filled with hot particles and various processes occurred, leading to the formation of atomic nuclei for the first time. Modern space science, based on mathematical descriptions, supports the existence of atomic nuclei during the Big Bang event. Calculations indicate that there were numerous atomic nuclei, with hydrogen being the most abundant, followed by helium and a small amount of lithium. Oxygen, which is essential for water formation, was likely absent during this time. Approximately 100 million years after the Big Bang event, stars emerged in the universe. The inner part of stars reaches extremely high temperatures, earning them the name "superheated furnaces." Scientists believe that when stars undergo supernova, the resulting explosion scatters elements into space, leading to the formation of complex elements from molecular nuclei. These complex elements include carbon, nitrogen, and oxygen. Water is formed through the combination of hydrogen and oxygen, a process likely completed after the Earth cooled and developed its atmosphere.

Water is found on Earth in various natural water sources and structures created by both nature and humans. Natural sources include the sea, clouds, rainfall, snow and glaciers, wetlands, rivers, and waterfalls, as well as underground water reservoirs and hot springs. Human-made water structures include lakes, ponds, reservoirs, water bodies, wells, step wells, tanks, canals, dams, barrages, tube wells, and stop dams, among others. Indian hydrologists believe that the Earth's water cycle is significantly larger than the planet itself. According to their estimations, approximately 1 billion, 91 lakh, 85 thousand, and 10 years ago, when the Earth took its present form, the world's expansion was approximately 50 crore yojans. At that time, there were seven continents, each more than twice the size of the others, arranged as a series of concentric circles and surrounded by seven seas of varying sizes. This is an Indian belief regarding water.

Modern scientists have estimated the quantity of water on Earth using figures and statistical methods. Statistical methods, primarily based on figures, are commonly used by hydrologists and managers. According to these estimates, the total amount of water on Earth is approximately 13,100,000 cubic kilometers.

THE NATURE OF WATER

Water, due to its inherent properties, exhibits variability in its distribution and quantity across different locations. It undergoes changes over time and will continue to do so. This variability is evident in the form of warm and cold water currents that flow in the oceans. Water exists in different states within the atmosphere, including vapor, liquid, and solid (ice). Additionally, it becomes incorporated into the Earth's depths as magma and can resurface through volcanic eruptions, often carrying various minerals. These characteristics are intrinsic to water's nature.

SYNONYMS FOR WATER

The 27 main synonyms for water are as follows:

Aapah, Va, Vaari, Salil, Jala, KamalamPayah, Keelala, Toyam, Amrit, Jeevan, Bhuvan, Vanam, Kabandham, Udakam, Paathah, Puskar, Sarvatomukham, Ammah, Arnah, Paaniya, Neer, Ksheeram, Ambu, ShambaramMeghapushyam, Evam Dhanarajah.

FORMS OF WATER

Water can be divided into three forms:

H₂O (H₂O, Hydrogen Oxide) - Water is formed by the combination of two hydrogen atoms with one oxygen atom. We use this form of water as drinking water, commonly referred to as ordinary water. Ordinary water boils at 100 degrees Celsius and freezes at or below zero degrees Celsius. This state of water is essential for the existence of living beings as it serves various daily needs, including agriculture.

D₂O (D₂O, Deuterium Oxide) - This water is approximately 11% heavier than ordinary water due to the presence of deuterium, a heavier isotope of hydrogen. It is commonly used as a coolant in nuclear reactors. Heavy water freezes at 3.8 degrees Celsius and boils at 101.4 degrees Celsius.

T₂O (Tritium Oxide) - This type of water is commonly referred to as superheavy water. It is approximately 33% heavier than ordinary water. Superheavy water freezes at 9 degrees Celsius and boils at 104 degrees Celsius.

Water is an essential compound for all living beings and is abundant on the Earth's surface

Information and Properties

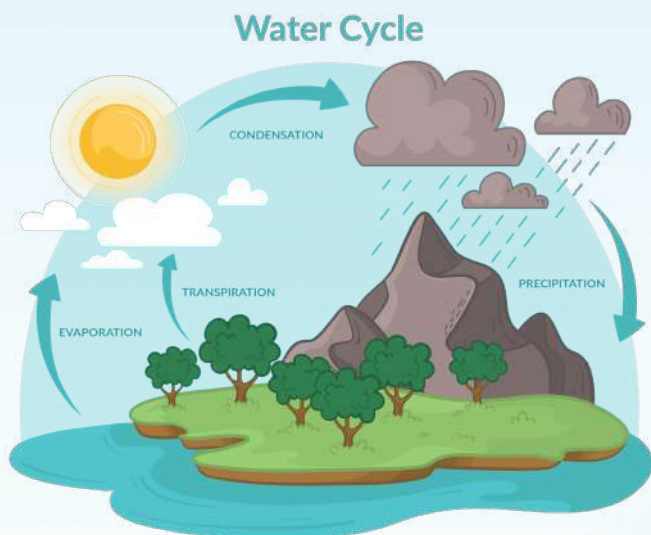
IUPAC Name:	Oxiden
Alternative Names:	Aqua, Dihydrogen Monoxide, Hydrogen Hydroxide
Chemical Formula:	H ₂ O
CAS Number:	7732-18-5
InChI:	Inchl=1/H ₂ O/h1H ₂
Molar Mass:	18.0153 g/mol
Density and State:	0.998 g/cm ³ (liquid at 20°C, 1 atm) 0.917 g/cm ³ (liquid at 0°C, 1 atm)
Melting Point:	0°C (273.15K) (32°F)
Boiling Point:	99.974°C (373.124K) (211.95°F)
Specific Heat Capacity:	4.184 J/(g·K) (liquid at 20°C) 74.539 J/(mol·K) (liquid at 25°C)

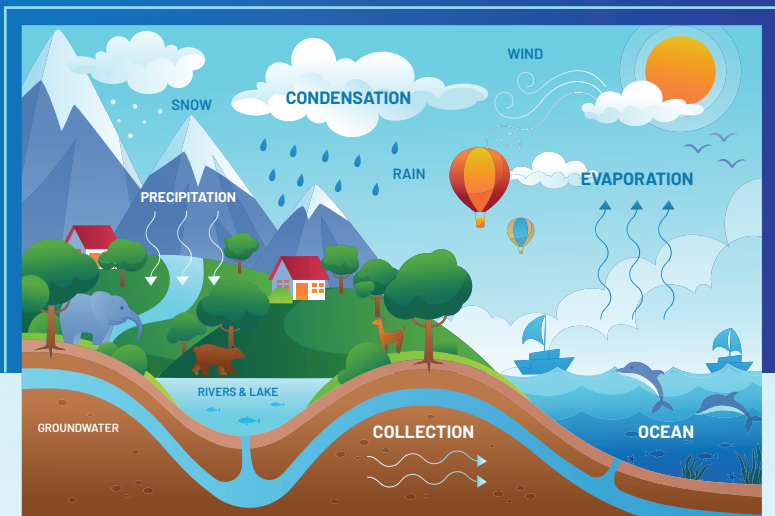
Source: Wikipedia

WATER CYCLE

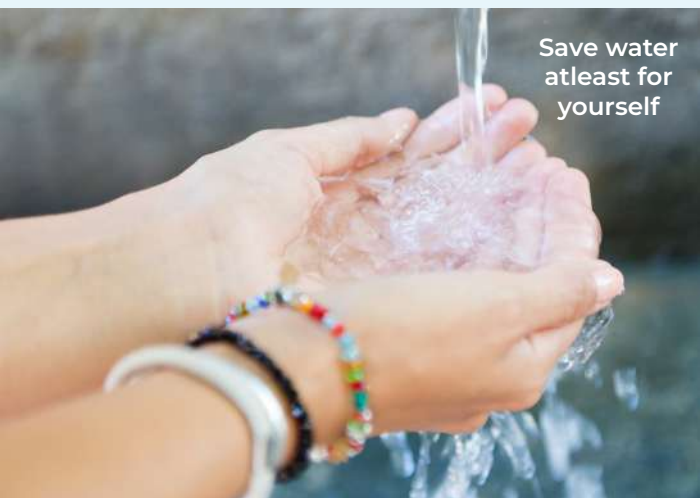
The natural water cycle is nature's-controlled water management system, skillfully established since ancient times to fulfill various responsibilities on Earth. There are numerous pieces of evidence and indications on Earth that help us understand this cycle. Its components work together to create favorable conditions for life in ecosystems while shaping the Earth's physical, chemical, and biological processes. The natural water cycle acts as a continuous journey for water, passing through the oceans, atmosphere, and land. The transformation of water into vapor shows that, regardless of its initial pollution, it becomes pure when it changes state. Even when it falls as rain, it remains pure, but its purity can be compromised upon reaching the Earth's surface.

The balance of the natural water cycle is remarkable, as the amount of water that evaporates from the Earth and reaches the atmosphere is equal to the amount of water that returns to Earth as rainfall. The water vapor in the atmosphere remains constant and is eventually returned to the seas through rivers and groundwater. This arrangement has been operating for millions of years and will continue to do so, although climate change may impact its different states, the total quantity and journey of water remain unchanged. It shows that water travels approximately 15 kilometers high in the atmosphere and penetrates up to about one kilometer deep into the upper layer of the Earth. This journey is designed by nature to fulfill various responsibilities. While it provides conditions for life in some places, it also brings about physical or chemical changes in others. The natural water cycle may seem simple to the common person, but in reality, it involves numerous complex processes. It cannot be adequately represented as a linear sequence. There are multiple branches within it that exist at the local, regional, and continental levels and continue to operate continuously. It functions as a complex system where water flows continuously, fulfills its responsibilities, and reaches its outcomes. It has no beginning or end.





The flow of rivers is an essential part of this journey. It is true that the quantity of water present in the water cycle varies depending on time and weather, but its total amount remains constant. Its distribution also constantly changes at the local, regional, and continental levels. There are similarities and differences in the water flowing on the surface and the water flowing beneath the ground. When clean and pure water falls on the Earth in the natural water cycle, some of it joins the river system. In this journey, some impurities mix with the water, and it carries them along and transports them forward. The river system, including impurities (suspended and solid), deposits that water back into the sea. The natural water cycle performs this function to keep the Earth clean and habitable for living organisms. It disposes of all waste into the sea as part of this system. Even in adverse conditions, it provides protection to life.



Save water
atleast for
yourself

It is estimated that approximately 13,100 million cubic kilometers of water are included in the natural water cycle. This partnership is responsible for maintaining the cleanliness of the Earth and providing the foundation for life. Saltwater primarily transforms into vapor and converts into clean water, which then sustains life on continents.

AVAILABILITY OF

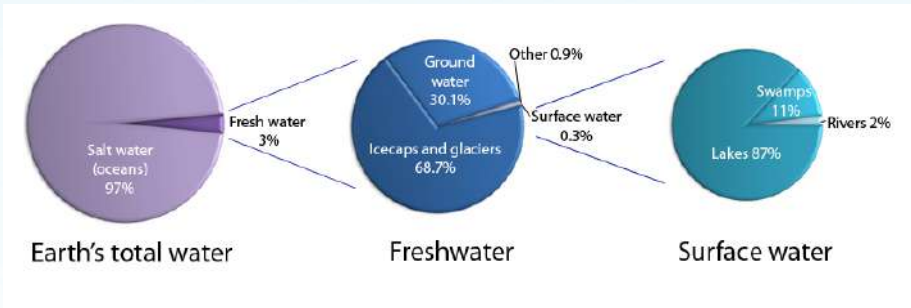
AVAILABILITY OF WATER ON EARTH

In India, the average annual rainfall is approximately 1,100 millimeters. The country has a freshwater availability of about 1,869.35 cubic kilometers per year, which includes water collected from glaciers, lakes, rivers, and other natural sources. Additionally, India has a significant groundwater availability of 435.6 billion hectare meters per year. It's important to note that approximately 70% of the Earth's surface is covered with water. However, out of the total water on Earth, 97% is found in the oceans and seas, and it is not suitable for drinking due to its salinity. Only 3% of the Earth's water is freshwater. The distribution of freshwater is as follows: approximately 68.7% is locked in glaciers, 30.1% is found in groundwater, 0.3% is available on the surface, and 0.9% is obtained from other sources. Out of the 0.3% available on the surface, 87% is found in lakes, 2% is in rivers, and 11% is contained in moist soil.

PER CAPITA WATER AVAILABILITY IN INDIA		
Year	Population [millions]	Per capita water availability [cubic meters]
1951	361	5177
1955	395	4732
1991	846	2209
2001	1027	1820
2025	1394 [estimated]	
2050	1640 [estimated]	1140

Source: Central Water Commission

Distribution of water on earth



USES OF WATER



The Five Elements obtained from Nature, with water being one of them, are essential for life. Without water, our survival would not be possible, which is why water is often referred to as the essence of life. From morning till night, we rely on water for various purposes, such as brushing our teeth, bathing, cleaning clothes, using the restroom, maintaining household cleanliness, washing vehicles, cooking, washing dishes, irrigating crops, industrial processes, and providing water for animals to drink, among many other uses. If we observe closely, water is an integral part of our daily lives. In India, the per capita availability

of water is 140 liters, while the World Health Organization recommends a minimum of 200 liters of water per day for an individual. It is worth noting that approximately 85-90% of villages in India rely on groundwater as their primary water source.

Water constitutes about 60-70% of our body composition. Our bodies excrete water through sweating and urination at an average rate of approximately 250 milliliters per hour. Renowned scientist and science writer J.B.S. Haldane once humorously remarked to the Archbishop of Canterbury that he was also 65% water. His statement holds truth as a significant portion of our brains, blood, and heart consists of water.

In the ocean, almost 95-97% of the total weight of living organisms is composed of water. A human fetus at one month of development contains 93% water. An egg is composed of 74% water, while approximately 75% of meat is water. Most terrestrial plants on Earth contain about 40% water, whereas aquatic plants can contain up to 90% water. Cucumbers, for example, are comprised of about 95% water.

The importance of water in our daily lives can be easily understood when we consider the numerous activities throughout the day that require its use.

LIFESPAN OF A SINGLE DROP OF WATER

Name of the water body	Duration of persistence (on average)
In the atmosphere	Up to 8 days
In the ocean	Up to 2,500 years
In glaciers	Up to 9700 years
In fresh water lakes	Up to 17 years
In rivers	Up to 16 days
In soil moisture	Up to 1 year
In ground water	Up to 1400 years

Source: Central Ground Water Board



THE IMPORTANCE OF WATER

In the development of Civilizations

The importance of water has been recognized for centuries, dating back to ancient civilizations. From the development of early societies to the present day, water has played a vital role in our lives. The Mesopotamian civilization thrived along the banks of the Tigris and Euphrates rivers in present-day Iraq, harnessing the power of these waterways for agriculture and trade. Similarly, the Indus Valley Civilization flourished along the banks of the Indus River, where water was crucial for sustenance and irrigation.

These ancient civilizations provide significant evidence of the fundamental importance of water. The remains of Mohenjo-Daro and Harappa, two prominent cities of the Indus Valley Civilization, further highlight the reliance on water for their existence and prosperity. Their advanced water management systems and infrastructure demonstrate the understanding of the significance of water in their daily lives.

Even in the modern age, the pattern of major cities being located near rivers continues. Cities like London along the Thames River, Washington D.C. along the Potomac River, Prayagraj (formerly Allahabad) along the Ganges River, and Delhi along the Yamuna River exemplify this trend.

In Religious beliefs

The significance of water is mentioned in ancient Hindu scriptures, such as the Vedas and Puranas. In the Rigveda, water is described as one of the five great elements, while the Atharvaveda refers to water as the sustainer of life. Hinduism considers Varuna as the deity associated with water, and according to religious texts, many avatars of God resided in water. An example of this is the emergence of nectar (Amrita) during the churning of the ocean (Samudra Manthan).

The association of water with religious rituals is deeply rooted in ancient traditions. Water bodies are worshipped and cared for. The Manusmriti states that contaminating clean water intended for public use should be punished with death. The importance of water is also mentioned in other religious texts, including the Vedas, Guru Granth Sahib, Quran, and Bible. It is common to find wells and ponds constructed near religious sites. In Jewish and Christian beliefs, the presence of ponds near gurudwaras (Sikh temples) symbolizes the separation of Earth from Heaven by God.

In the Production of Goods

The production of various goods requires a significant amount of water, ranging from a few liters to millions or even billions of liters. Water is utilized in the manufacturing of vehicle components, their preparation, petroleum refining, the paper industry, and the sugar industry. It is estimated that approximately a hundred thousand liters of water are consumed in the production of a single car. The purification process for petrol requires millions of liters of water. As a result of the water-intensive purification process, Gulf countries primarily export crude oil. Industries that employ water for cooling purposes also contribute to maintaining normal temperatures in colder regions. The continuous flow of water from industrial drains is indicative of its extensive usage in industries. Even the construction of a house requires hundreds of thousands of liters of water. The production of books, writing on registers, and reading newspapers also consumes a significant amount of water.

In Agriculture

The highest consumption of water in the world occurs in agriculture. In addition to using water from rivers for irrigation, groundwater is extensively utilized in agricultural practices. It is estimated that approximately 85% of groundwater is used for irrigation purposes.

The area of agricultural land irrigated using groundwater has significantly expanded over the years, from 6.5 million hectares in 1951 to nearly 60 million hectares by the year 2020.

The introduction of chemical fertilizers and pesticides in agriculture has contributed to a significant increase in water consumption. According to a survey, the production of one kilogram of rice requires approximately 30,000 liters of water, while producing one kilogram of wheat, which is

commonly used for bread, requires around 1,500 liters of water. Water consumption is also substantial in crops such as sugarcane. Consequently, many countries avoid cultivating water-intensive crops.

In Electricity Production

When we work on a computer, watch a program on television, or use air conditioners, coolers, and fans, we rarely pay attention to where the electricity is coming from. However, it's important to note that electricity production often relies on water as a resource. Electricity is generated by harnessing the flow of water in large dams, which are constructed to control the water flow in rivers. Some of the major dams in India include Bhakra Nangal, Tehri, and Pong. Zambia has the Kariba dam, Brazil has the Tucuruí dam, and the United States is home to the Hoover Dam. These dams play a crucial role in generating electricity from water. Additionally, smaller hydroelectric projects are also being implemented to generate electricity from water. In India, there are approximately 14 million acres of land dedicated to dams and reservoirs for water control.

The Search of Water

Even in the era of the information revolution and technology, when scientists contemplate the possibility of living on other planets or the existence of life similar to Earth, the search for water always takes precedence. Scientists strive to gather information that could make life feasible in those places. However, to date, no scientist in the world has obtained conclusive evidence. It is speculated that there may be some ice present on the planet Mars, but solid proof is still lacking. The existence of methane gas on certain planets is believed to indicate the possibility of life. Billions of dollars have been invested in the quest for water, which essentially equates to the search for life itself.

Cry for Water

Examples of water scarcity can be observed in Rajasthan, Delhi, Shimla, Latur, and Chennai. The true value of water becomes apparent to those women who carry water on their heads for several kilometers. Those who stand in line for hours, waiting for water tankers to arrive in their neighborhoods, understand the significance of water. No one can explain the importance of water better than a farmer. Farmers in Rajasthan, who have to invest a significant amount of money to deepen their wells, can vividly describe the water situation. And who can comprehend it better than those who diligently collect each raindrop in tanks and wells?

In 8 out of 20 river basin areas in the country, there is a water shortage. According to estimates, the groundwater level is rapidly declining in 362 districts across the country.

The Current Situation of Water

Discussions regarding water are taking place both in the country and around the world. Some view it as a potential cause for the third world war, while others seek to profit from it. Currently, water is transitioning from being a communal resource to a private asset. Large companies, driven by profit, are eyeing the water business. The water trade today is worth billions of rupees. Although water from rivers is a natural resource, there is a concerning possibility that it may become a commercial commodity in the future. Water is being sold at higher prices than milk, indicating that water scarcity will lead to increased costs in the future.

It is believed that farmers will suffer the most after water privatization. When water falls into private hands, there will be a shortage of water available for crop irrigation. Farmers will no longer have rights over rivers, canals, or adjacent streams flowing through their fields. They will be unable to access water from these sources for irrigation or even for drinking. They will also lose the ability to dig wells on their own land. These are not hypothetical scenarios but potential harsh realities of the future. Unless the current situation improves, society will face these challenges.

Today, we are compelled to question the chaos surrounding water and its privatization. What does it truly entail? The privatization of water is, in essence, an encroachment on our collective rights by multinational companies. These rights belong to all of us who hold sovereignty over water and all other natural resources. In the future, the water market is expected to become the largest because markets are driven by demand. The privatization of water is not a small-scale endeavor; rather, it involves complex and interconnected strategies.

Controversies Surrounding Water

When a commodity becomes scarce or its scarcity is anticipated, its significance naturally increases. Water has become such a commodity, and the echoes of controversies surrounding it are now heard everywhere. Disputes over drinking water and water for crop irrigation are widespread. These conflicts occur not only between countries but also within national boundaries. Disputes over water have emerged even among communities, and sibling rivalries over water are frequently heard and reported. These water-related disputes take various forms.

Controversies over water from rivers like the Cauvery and Yamuna continue to persist in different states. By examining an example, we can understand the potential future water disputes. Delhi receives water supply from the Ganga Canal in Muradnagar, upon which millions of farmers rely for irrigation of their crops. If farmers are one day deprived of canal water or if its supply is reduced, it can lead to a serious dispute. Farmers from western Uttar Pradesh would demand water from the Ganga Canal, while residents of Delhi would demand drinking water for themselves. Additionally, the continuous decline in water levels in rivers results in reduced water in canals. Water disputes also arise in small settlements and colonies. In rural areas, physical fights over irrigation water have become common, to the extent that these disputes sometimes result in loss of life. Incidents of mass killings have occurred in Muzaffarnagar, Bijnor, and Saharanpur due to water disputes. In Eastern Uttar Pradesh, incidents of water filling at gunpoint are frequently reported. Water disputes are also frequently observed at the international level.



Water is Becoming a Weapon

The present era is dominated by global warming. Developed countries strive to progress further, while developing nations work hard to achieve development. However, impoverished countries are struggling to meet their basic needs. Each country is some have turned water into a weapon. Lakes and dams are being prepared as "water bombs," and engaged in increasing its arsenal for the sake of security, and water trade-offs are being made in exchange for river water. Countries located at higher altitudes than sea level are constructing artificial reservoirs to control the flow of rivers, posing a continuous threat to countries at lower elevations. Certain multinational companies, enriched with water resources, provide loans to impoverished and developing countries. In return, they attempt to establish control over the rivers and other natural water sources of those countries. They seek to exert dominance over water and other natural water sources as a way to compensate for the loans given. A state that has depleted its groundwater or reached a critically low level becomes dependent on water from other states. It depends on the governance of that state how it negotiates the trade of its water resources.

What can we do?

We need to take action starting today to conserve water. We must not wait until we become seriously concerned about water scarcity. Water availability on Earth, in whatever form it may be, is continuously diminishing. Groundwater levels are depleting, rainfall is decreasing, and even river water is becoming scarce. In such circumstances, we need to seriously consider saving water. To carry out water conservation efforts, we should work on two levels. Firstly, we need to minimize our water usage, and secondly, we need to preserve and protect water resources.



There are various ways in which we can contribute to water conservation. Some efforts can be made at an individual level, while others require collective action on a larger scale, starting from our homes. At the household level, we can save thousands of liters of water daily by making changes in our daily habits. Implementing rainwater harvesting techniques in our homes can also contribute to water conservation. In rural areas, natural water sources can be rejuvenated, and rainwater can be conserved. Water-saving measures are provided under environmental conservation activities, with specific water dimensions specified for each household. Activities aimed at water conservation have been designed for every household under green house programs.

Green House

The Green House program, implemented under environmental conservation activities, is an initiative through which every family can contribute to environmental and water conservation. It has been determined that each family member can lead an environmentally friendly life through small efforts. This nationwide program allows every family member to play a supportive role. The Green House program is based on five principles known as 5-4-3-2-1. These principles encompass five points for water conservation, four points for soil conservation, three points for energy conservation, two points for tree conservation, and one point for biodiversity conservation. To actively participate in the program, we need to perform the following tasks:

Water Conservation

1. Bathing with a bucket of water.
2. Half a glass of water.
3. Conservation of RO/AC water.
4. Preventing water leakage from pipelines.
5. Using mist or aerator for water.

Soil Conservation

1. Waste Segregation
2. Eco-Bricks
3. Kitchen Garden
4. Composting

Energy Conservation

1. Turn Off Unused Appliances
2. Use LED Lights
3. Utilize Alternative Energy Sources

Tree Conservation

1. Planting Greenery
2. Establishing Mini Forests

Wildlife Conservation

1. Provision of Water for Birds and Animals



Rainwater Harvesting

CHAPTER TWO



Over the past two decades, rainwater harvesting has emerged as a prevalent subject of discussion and practice between the government and society. So, what is rainwater harvesting? It is the process of preventing the wastage of water obtained through rainfall and efficiently channeling it to the groundwater through scientific methods. Rainwater harvesting is carried out in both urban and rural areas, utilizing different techniques.

The utilization of rainwater harvesting varies based on different circumstances. In hilly regions and areas like Rajasthan, where water scarcity for drinking purposes persists, rainwater is collected and stored underground or in tanks to be used throughout the year. In regions like western Uttar Pradesh, where there is currently no scarcity of drinking water, rainwater is directly channeled to the groundwater.

The entire world is facing a water crisis, and the only solution that seems evident is rainwater. If we can conserve rainwater, we can immediately and permanently address the problem. Different methods are being adopted for rainwater conservation across the country, but there are also some misconceptions associated with it. In such a scenario, it becomes our duty to present the correct facts and arguments to society. We should not get trapped in complex terminology, scientific language, and intricate procedures related to rainwater conservation. Our effort should be to communicate the



correct message of rainwater conservation to the general public. Rainwater conservation is only about rainwater and not any other type of water (kitchen, sewage, bathroom, industry, etc.). It is a technique to bring rainwater to groundwater. Sending any kind of polluted water to the groundwater is an offense because it will contaminate it. Rainwater conservation is purely a scientific method. It is one technique. Through this booklet, we are providing information about rainwater conservation and the correct ways to implement it. Rainwater is a precious gift given by nature to every living being on Earth. It is worthy of worship, and we should respect it.

In rural areas, we already practice rainwater conservation through ponds, but in towns and cities where the quantity of ponds is low or negligible, rainwater conservation can only be achieved through rainwater conservation techniques. We need to increase awareness where necessary, and if someone wants to undertake the work of rainwater conservation through these techniques in their own homes, then they will need to form a team responsible for designing and constructing the entire structure.

Everyone should practice rainwater conservation in their homes, shops, and other places. Encourage your neighbors and colleagues to participate in this work. Engage in discussions with people about the benefits of rainwater conservation. Make people aware of the advantages of rainwater conservation. Provide them with complete and accurate information about it. Convey to the nation the importance of preserving rainwater for the future and the understanding that we should ensure that the amount of water we extract from the earth is proportionately replenished in the groundwater. By conserving rainwater, you can raise the groundwater level and store it for daily needs as well. Until we have your support in this campaign, the method of rainwater conservation cannot progress. We must always remember that a pot is filled drop by drop, but it can also be emptied drop by drop.

Rainwater Conservation: Some Tips

Save every drop



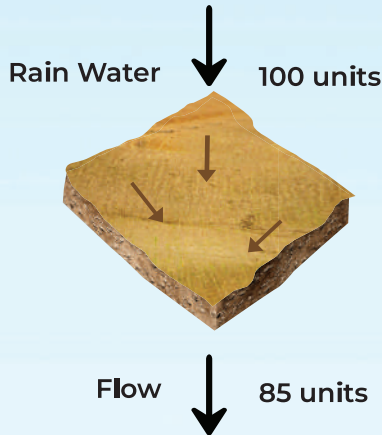
What is wasted water?

Wasted water refers to water that falls as rain on a surface and flows away from the catchment area without being utilized. The water runoff from rooftops and various types of surfaces can be controlled and managed. The quality of the collected runoff water depends on the

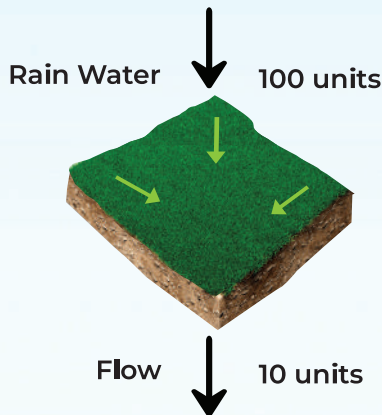
characteristics of the surface it comes into contact with. For instance, approximately 85% of the rainwater that falls on tiled roofs flows away as wasted water. In contrast, only 30% of the rainwater that falls on unpaved surfaces or surfaces covered with wood or grass is lost as runoff.

The remaining water tends to be retained on the surface and gradually infiltrates into the ground. Paved surfaces, although not completely impermeable, allow for less infiltration, resulting in around 70% of the water flowing away as wasted water. In terms of water quality, wasted water can be categorized into three types: water flowing from rooftops, water flowing from paved surfaces, and water flowing from unpaved surfaces. Water flowing from unpaved surfaces may contain bacteria or other contaminants. Hence, if water is to be collected for drinking purposes, it is advisable to focus more on collecting water from rooftops, where the quality is likely to be better.

Water flowing over plain surface



Water flowing over grass covered surface

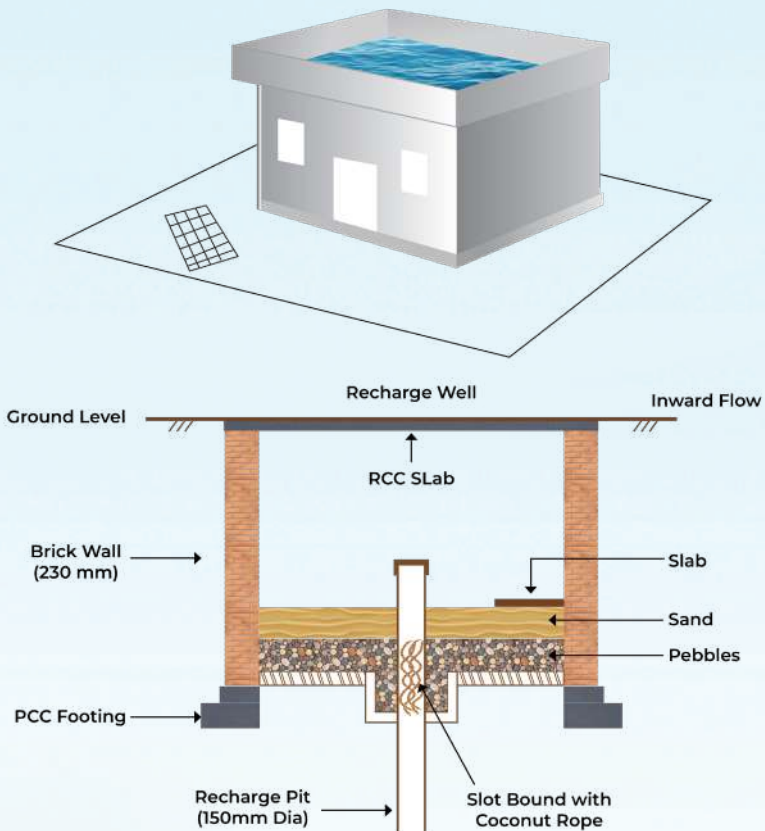


What is recharge well?

A recharge well helps replenish the groundwater level.

- The depth of a recharge well should not be more than 5 meters below the water table.
- The minimum diameter or circumference of a recharge well should be 0.1 meters, and its edges should be filled with gravel.
- There should be a suitable pit above the recharge well.
- The well should be large enough to capture and retain the water flowing from an area of 500 square meters.
- For a 500 square meter area, a recharge well has the capacity to store 3-4 cubic meters of water.
- There should be a recharge borehole inside the recharge well so that the collected water can percolate into the groundwater aquifer.

Position and Description of Recharge Well near a Building



Rainwater Conservation in Cities

To implement rainwater conservation in ten cities, a specific technical process is followed. The process involves collecting rainwater from roofs, courtyards, or open areas using pipes, which then directs the water to a designated location for underground storage. Before the water is sent underground, a pit known as a recharge well is excavated based on the area of the rainwater conservation site, with dimensions of length, width, and depth. Multiple pits can be constructed depending on the size of the conservation area. Boring is conducted between these pits.

The width of the boring is determined based on the expected water flow on the surface. A collection chamber, a small pit, is created near the recharge well. The rainwater first enters the collection chamber, allowing any accompanying debris to settle, ensuring that only water enters the recharge well.

During the boring process, a pipe is inserted, and it is positioned at a specific height, typically three to four feet above the groundwater level, on a specialized layer. The lower end of the pipe, coming out of the recharge well, is provided with a hole of about two feet in length and half an inch in diameter. This hole is covered with a plastic mesh, and a jute rope is tightly wrapped over the mesh. The pipe extends above the pit, and a small cap with small holes is placed on the open upper end of the pipe to allow for air circulation inside the pipe.

The pit where the boring pipe is placed is surrounded by a layer of bricks or stones, approximately six inches thick, from all sides. Additionally, a layer of gravel, about six inches thick, is added, followed by a final layer of sand, also about six inches thick. These layers, totaling eighteen inches, serve as filters for the rainwater before it reaches the groundwater. This ensures that the water entering the pit undergoes thorough filtration before reaching the groundwater reserves.

Traditional Methods of Rainwater Conservation Practiced in Different States of India

Uttar Pradesh	Johad (Check Dams), Talab (Ponds), Kua (Wells)
Bihar	Aaha Phyne (Diversion-Cum Storage System)
Gujrat	Baavdi (Stepwells), Jheel (Lakes)
Tamil Nadu	Eri (Tank)
Maharashtra	Bandhara (Embankments), Phad (Small Dams)
Rajasthan	Kund (Underground Tank), Kui (Well), Johad (Check Dam), Anicut (Diversion Weir)
Ladakh	Zing (Small Tank)
Meghalaya	Channeling Water through Bamboo Pipes

Source: Water Foundation

Why is Rainwater Conservation?

Rainwater conservation is necessary in India due to the country's average annual rainfall of around 1200 millimeters, which occurs within a short span of 100 hours. This intense and rapid rainfall results in a significant amount of rainwater being wasted. Only a small portion of the rainwater is able to percolate into the groundwater.

The groundwater reserves are extensively utilized by humans for agriculture, industries, and domestic purposes. As a consequence, the groundwater level is continuously declining. Moreover, farmers consistently require more water for high-water-consuming crops. In such circumstances, rainwater conservation becomes crucial.

Precautions for Rainwater

- Ensure that the borewell pipe is positioned properly, at least three to four feet above the groundwater level.
- When practicing rainwater conservation, it is important to collect only rainwater and allow it to recharge the groundwater. Avoid mixing it with sewage, kitchen waste, industrial waste, or any other form of contaminated water.
- Design and dimension the rainwater harvesting system according to the area's annual rainfall and the capacity of the pipes and borewell. Seek guidance from experts in the field for proper implementation.
- Securely wrap a plastic mesh and jute rope around the borewell pipe to prevent debris from entering.
- It is advisable to consult individuals with technical knowledge and experience in rainwater conservation to ensure proper implementation. Even a small mistake can have adverse consequences, so seeking expert guidance is important.

Where should Rainwater Conservation be done?

- Rainwater conservation should be prioritized in areas where the groundwater level is eight meters or more below the surface.
- Implementation is particularly crucial in regions experiencing a continuous decline in the groundwater level.
- It is also recommended in locations where there is substantial water percolation from below the ground.
- Parks and clean roadsides, as these areas facilitate the easy accumulation of rainwater.

Benefits of Rainwater Conservation:

- The groundwater level rises.
- It reduces the level of pollution in the groundwater.
- It reduces the risk of floods.
- Minimizes the impact of drought.

- Rainwater, when conserved, doesn't get wasted by flowing into drains, canals, rivers, or the sea.
- When the groundwater level increases, the electricity consumption for pumping water through tube wells, bore wells, submersible pumps, etc., decreases.
- There is a decrease in soil erosion.
- It provides relief from waterlogging issues.
- The occurrence of waterborne diseases decreases due to waterlogging.
- Water is available for drinking purposes.

Important Considerations:

- For buildings and other locations where rainwater conservation systems are installed, the following points should be taken into account:
- The roof should be thoroughly cleaned before the onset of the rainy season.
- After the rainfall starts, allow the initial five to ten minutes of water to flow out, as it may carry environmental pollutants.
- Clean the filters installed on pipes to prevent dirt accumulation before the rainy season.
- If rainwater conservation is implemented in a park or open area, avoid the use of pesticides and chemical fertilizers.

Rainwater Conservation Area

A rainwater conservation area is a location where rainwater directly falls. This can include the solid roof of a house, the courtyard of a home, an open space, a garden, a terrace, or any other suitable area.

How to calculate the quantity of collected rainwater?

To determine the quantity of rainwater collected in a rainwater conservation area, you will need to know the amount of rainfall at that location (usually measured in millimeters, centimeters, or inches) and the area of the space where rainwater is being collected

For example:

Let's assume the rainwater conservation area has an area of 100 square meters, and the average rainfall received at that location is 600 millimeters (24 inches) per year.

Capacity of the rainwater conservation area = Area of the space (100 square meters) × Average rainfall (0.6 meters or 600 millimeters/24 inches)

The amount of water obtained in the rainwater conservation area = Capacity of the rainwater conservation area × Collected rainfall

= 100 square meters × 0.6 meters = 60 cubic meters (60,000 liters) of water saved.

Rainwater Harvesting Area (in square meters) × Amount of Rainfall (in millimeters) = Quantity of Collected Rainwater (in liters)

Scientific Evaluation of Rainwater Availability from Rooftop Collection

Rainfall (mm)	100	200	300	400	500	600	800	1000	1200	1400	1600	1800	2000
Roof top area (Sq m)	Harvested Water from Roof top (cu m)												
20	1.6	3.2	4.8	6.4	8	9.6	12.8	16	19.5	22.4	25.6	28.8	32
30	2.4	4.8	7.2	9.6	12	14.4	19.2	24	28.8	33.6	38.4	43.2	48
40	3.2	6.4	9.6	12.8	16	19.2	25.6	32	38.4	44.8	51.2	57.6	64
50	4	8	12	16	20	24	32	40	48	56	64	72	80
60	4.8	9.6	14.4	19.2	24	28.8	38.4	48	57.6	67.2	76.8	86.4	96
70	5.6	11.2	16.8	22.4	28	33.6	44.8	56	67.2	78.4	89.6	100.8	112
80	6.4	12.8	19.2	25.6	32	38.4	51.2	64	76.8	89.6	102.4	115.2	128
90	7.2	14.4	21.6	28.8	36	43.2	57.6	72	86.4	100.8	115.2	129.6	144
100	8	16	24	32	40	48	64	80	96	112	128	144	160
150	12	24	36	48	60	72	96	120	144	168	192	216	240
200	16	32	48	64	80	96	128	160	192	224	256	288	320
250	20	40	60	80	100	128	160	200	240	280	320	360	400
300	24	48	72	96	120	160	192	240	288	336	384	432	480
400	32	64	96	128	160	192	256	320	384	448	512	576	640
500	40	80	120	160	200	240	320	400	480	560	640	720	800
1000	80	160	240	320	400	480	640	800	960	1120	1280	1440	1600
2000	160	320	480	640	800	960	1280	1600	1920	2240	2560	2880	3200
3000	240	480	720	960	1200	1440	1920	2400	2880	3360	3840	4320	4800

Source: Central Underground Water Board



An underwater photograph showing a large amount of plastic waste, including bottles and bags, floating in the water. The sun is visible through the surface, creating a bright, hazy light. The water is a deep blue-green color.

Water Pollution

CHAPTER THREE

When the clean water we use for drinking purposes becomes unsuitable for consumption due to contamination, it is referred to as polluted water. Polluted water contains dissolved chemicals and poses a risk to health when consumed, leading to potential health hazards. Polluted water is often discolored, odorous, and may have a distinct taste. Water pollution can be categorized into two types: visible pollution that can be observed with the naked eye, and invisible pollution that cannot be detected visually. To assess the invisible pollutants, scientific methods and tools are required.

Water pollution can be described as an increase in the level of contamination in lakes, rivers, oceans, or any other water source, whether caused by natural or human factors. As pollution levels rise, the pollutants gradually dissolve and mix with the water. Such polluted water is harmful to health.

Visible Water Pollution

Visible water pollution refers to the pollution that can be observed with the naked eye. This type of pollution is characterized by contaminants that are visible in the water, such as floating debris, plastics, and other waste materials. An example of visible water pollution is the blackened water flowing through drains in cities, which often contains untreated domestic wastewater and liquid waste from industries.

These polluted drains eventually merge with rivers and ultimately reach the ocean, contributing to the contamination of water bodies.

In addition to industrial and domestic waste, there are also cultural and religious practices that contribute to water pollution. Immersing statues, ashes, and the bodies of the deceased, as well as discarding fruits and flowers in rivers, are examples of how religious activities can impact water pollution.

Invisible Water Pollution

Another form of water pollution is the invisible pollution that cannot be seen with the naked eye but can be highly dangerous. This type of pollution is primarily caused by the use of chemical fertilizers and pesticides in agricultural practices, as well as the discharge of polluted water from households and industries. Invisible water pollution can also occur naturally. The danger lies in the fact that contaminated water, which may appear clear to the eyes, can still be harmful to human health.

Drinking water contaminated with invisible pollutants often lacks proper physical and chemical standards. Parameters such as Biological Oxygen Demand (BOD) and Chemical Oxygen Demand (COD), water hardness, fluoride and nitrate content, and abnormal pH levels can exceed the normal range, posing significant health risks. Consuming such polluted water can lead to the formation of kidney stones, weakened bones and teeth, and various gastrointestinal diseases. Unfortunately, people may remain unaware of these risks until the diseases become severe.

In addition to naturally occurring pollutants, water pollution can also result from the presence of heavy metals and pesticides in drinking water. The excessive use of hazardous elements like lead, chromium, cadmium, aluminum, as well as toxic substances like cyanide, arsenic, and pesticides in agriculture further contribute to the risks of water pollution, sometimes resulting in fatalities.

It is crucial that drinking water meets the standards set by organizations such as the World Health Organization and the Indian Standards Bureau. Any level of pollution that exceeds the specified standards in drinking water is a matter of concern and requires immediate attention.

Causes of Water Pollution

The causes of water pollution can be categorized into two main types: natural causes and human-induced causes.

1. Natural Causes:

Natural pollution in water occurs when pollutants naturally enter the water system. This can include minerals such as coal and petroleum from underground mines, toxic gases, soil erosion, leaves from trees, and animal waste, among others. Natural pollution tends to accumulate more in groundwater, but it can also spread to surface water bodies.

2. Human-Induced Causes:

Anthropogenic water pollution refers to pollution that is introduced into water bodies as a result of human activities. This type of pollution affects both surface water and groundwater. The increasing population is a significant driver of water pollution, as the growing demand for resources puts pressure on water sources. With the advent of modern living, water pollution has increased significantly. Some of the human sources of water pollution include:

By Industries

Did you know that industries play a significant role in spreading water pollution? Industries such as sugar mills, paper mills, textile mills, distilleries, and others utilize various chemicals and substances during the production process. Unfortunately, many of these industries continue to release their waste without proper treatment. It is common to find industries located near drains or rivers, as this facilitates easy discharge of their waste into water bodies. Consequently, the waste from these industries pollutes the clean water in drains and rivers. Over time, the polluted water seeps into the ground, resulting in groundwater pollution. It is important to note that polluting water is considered a punishable offense.



Water pollution also arises from the waste generated by meat industries located in major cities and towns. The slaughtering of animals in these industries produces waste that contributes to water pollution. Similarly, the use of chemicals in tanning hides leads to waste that is often discharged directly into drains or rivers. This waste not only pollutes the flowing water but also contaminates groundwater over time. These issues are prevalent in cities and towns where such industries are established. However, it is worth mentioning that industries are becoming increasingly aware of their impact on the environment and are making efforts to rectify their mistakes, particularly compared to the past.

By Agricultural Runoff

Have you ever considered the contribution of chemical fertilizers and pesticides used by farmers to water pollution? It is indeed true that only a portion of the chemical fertilizers and pesticides applied to crops by farmers is actually absorbed by the plants. The remaining percentage, approximately 75 percent for chemical fertilizers and around 90-92 percent for pesticides, is lost to the soil and the environment through various processes. These components eventually mix with water and find their way into drains, rivers, lakes, and ponds, thereby polluting the water within these sources. The pesticide particles, carried by water sources, can also reach groundwater.

Farmers have been using chemical fertilizers and pesticides in their crops for several decades. Over time, these chemical and pesticide particles gradually seep into the groundwater, posing a risk to water quality and environmental health.

By Medical Facilities

You would be surprised to know that when we fall ill or get injured in any part of our body, we are given medicines, bandages are applied, and injections are administered. However, when these medicines, injections, and bandages are discarded in open areas or thrown into drains, they contribute to water pollution. Today, due to contaminated food and water, diseases are increasing, which puts pressure on medical facilities to provide better healthcare services. In such situations, a large amount of waste is generated from medical facilities. Unfortunately, even today, some waste from medical facilities or homes ends up in empty spaces, one way or another. Improper disposal of waste by some medical facilities also becomes a cause of pollution.

In these medical facilities, broken thermometers release a significant amount of mercury. Mercury is a heavy metal. It may surprise you to know that just one gram of mercury can contaminate twenty acres of pond or lake water. If even a small amount of mercury enters the body, it can be life-threatening.

By Household Discharge

You may have observed that in our homes, various household activities such as cooking, bathing, and cleaning involve the use of products like soda, soap, phenyl, and other chemicals to kill pests and insects.

The waste generated from our homes contains a mixture of these chemicals and substances. This waste, when improperly disposed of, often finds its way into drains and sewage systems, eventually flowing into rivers. Consequently, household discharge becomes one of the primary causes of water pollution. Currently, approximately 80% of household discharge and 20% of industrial discharge flow into rivers through sewage systems.

By Dhobi Ghats

At the banks of rivers, canals, streams, and ponds, laundry areas known as dhobi ghats are set up by laundry workers for washing clothes. When clothes are washed at these dhobi ghats, the dirt from the clothes, as well as remnants of soap and other substances, flow and mix with the stagnant water, leading to water pollution. These impurities introduced by the dhobi ghats contaminate the water wherever it flows. The settled impurities in the stagnant water form a layer beneath the water's surface, hindering its ability to absorb moisture and gradually reducing the water retention capacity of the soil. This depletion of soil moisture over time has negative effects on the overall water balance.

Through the Use of Plastic

You may have observed the increasing use of plastic in various activities, such as carrying small items or transporting goods. When new items are brought into our homes, they are often wrapped or packaged in plastic. Unfortunately, plastic does not degrade for many years and tends to accumulate in drains and water channels. This trapped plastic interacts with the water, releasing its components and contributing to pollution. During rainfall, plastic can obstruct drains, causing water to accumulate in one place and promoting stagnant water, which further exacerbates pollution. Plastic pollution in oceans has become a severe global issue, leading to bans on certain single-use plastic products by many international organizations, including India. Plastic pollution is now a pressing problem that requires urgent attention.



Through Other Sources

Electronic waste is a rapidly growing issue both in our country and globally. Improper disposal of electronic waste can have detrimental effects on water quality, leading to water pollution. Water pollution can arise from various sources, including the dismantling of old ships, improper disposal of electronic waste in landfills, and the use of certain paints.

Types of Water Pollution

We can categorize water pollution into two types:

1. Surface Water Pollution

Surface water pollution is entirely humanmade. Any pollution that occurs in drains, rivers, lakes, and ponds is primarily caused by human activities. All the sources that gather or carry water on the surface, including the pollution that is increasing in them, are spread by human activities. Most of the rivers in the country are turning into dirty drains. Various studies have shown that these rivers contain high levels of pesticides and heavy metals. The water in village ponds is also getting polluted due to domestic waste and other forms of pollution. Lakes and wells are also suffering from the impact of pollution. Glaciers are also being affected due to climate change. Oceans have experienced a significant amount of pollution as contaminated water from rivers flows into them, and the movement of ships, their dismantling, and the leakage of petrol and other substances increase the pollution levels in the ocean waters.



Indian Standard Institution Standards for River Water Pollution

Parameters	Maximum limit
Ph	6-9
Chloride	6000 milligrams/liter
Dissolved Oxygen	40 percent
B.O.D	3 milligrams/liter
Oily substances	0.1 millimeter/liter
Cyanide	0.01 milligrams/liter
Lead	0.1 milligrams/liter
Chromium	0.05 milligrams/liter
Arsenic	0.2 milligrams/liter

Source: Central Pollution Control Board

HEAVY METALS

It is important for all of us to understand that the presence of heavy metals in water is highly dangerous to our health. High concentrations of various heavy metals in water can lead to different types of diseases.

Here are some of the associated diseases with specific heavy metals:

Heavy Metals	Disease Caused
Cadmium	Breathing difficulties, anemia, high blood pressure, kidney problems, nerve damage, damage to testes, and red blood cells
Chromium	Ulcers and lung cancer
Nickel	Skin diseases, lung and nasal cancer, dizziness, headaches, difficulty breathing, and vomiting
Lead	Significantly affects the kidneys. Joint pain, cancer, abdominal pain, weakened bones and teeth. It is also observed in 99% of fluorides
Iron	Severely affects the digestive process
Arsenic	Causes various skin-related diseases
Nitrate	Causes "blue baby" disease in children, and can also lead to cancer of the digestive system. Diarrhea occurs and digestive system excitement is generated
Copper Sulfate	Causes anemia in children, affects the liver, and causes irritability
Boron	Causes abdominal cramps and spasms
Zinc	Affects the sense of taste
Chlorine	Causes stomach cancer, respiratory and skin-related diseases
Chloride	Affects the sense of taste
Hard water	Can lead to kidney, stomach, and urinary bladder diseases

Source: Central Pollution Control Board

2. Groundwater Pollution

Groundwater pollution refers to the condition of water beneath the Earth's surface being contaminated by impurities. This pollution can occur in two forms: anthropogenic, which is caused by human activities, and natural. Even the impurities we introduce into surface water, such as plastics, garbage, and domestic waste, can affect groundwater. When polluted water flows through drains and rivers, it gradually infiltrates the ground. Over time, the impurities from the surface water accumulate in the groundwater, resulting in contaminated water emerging from hand pumps near rivers and drains.

Anthropogenic pollution in groundwater can be attributed to the long-term use of chemical fertilizers and pesticides by farmers. These substances gradually permeate the soil along with water, eventually reaching the underground water sources. Consequently, impurities like pesticides, which may not be visible in the groundwater, can still be found.

Natural groundwater pollution occurs due to the presence of arsenic and fluoride in the water beneath the Earth's surface. While these elements may naturally exist in some areas, their concentrations in groundwater can become elevated due to extraction processes.

Impact of Water Pollution

You will be surprised to know that water pollution affects the entire environment in various ways, not just humans but also animals, birds, and the soil. The effects of water pollution are as follows:

1. On Human Health

Water pollution has a significant impact on human health as we rely on water for our daily needs. In India, about 60% of diseases are waterborne. Contaminated water can lead to stomach-related illnesses like jaundice, diarrhea, and typhoid. It can also contribute to diseases related to heavy metals and pesticides, such as cancer, kidney problems, heart diseases, brain disorders, and cardiovascular diseases. Consuming polluted water can affect our digestion process and, in severe cases, even lead to death.

2. On Animals and Birds

Water pollution also affects animals and birds. Migratory birds from foreign countries have been known to die from drinking contaminated water. Instances of animal deaths due to polluted water have been reported, such as the case of 30 sheep in Meerut's industrial area. Fish and other aquatic organisms in rivers and lakes across the country are also experiencing the detrimental effects of increasing pollution. Studies have found traces of pollution in fish living in contaminated water. Animals and birds that consume polluted water often experience a decrease in reproductive capacity. Many bird species, including vultures, have become extinct or faced drastic population declines.

3. Aquatic Life

Rivers, drains, and seas are heavily contaminated, particularly due to plastic pollution. This poses a severe threat to aquatic organisms. Several small rivers in India have already lost fish and other aquatic creatures due to pollution. Marine life in the seas is also facing a crisis due to the escalating pollution levels.



4. Agricultural Soil

Farmers use groundwater for irrigation, which often contains significant amounts of pesticides. Studies have found banned pesticides in groundwater, and heavy metals have been detected in some areas. When polluted groundwater is used for irrigation, its components reach the crops through the soil, deteriorating its quality and impacting crop quality.



5. Food Crops

Water pollution has a significant impact on the quality and safety of food crops, fruits, and vegetables. Studies have detected traces of pesticides in vegetables, and excessive levels of pesticides have been found in various products, ranging from bottled water to noodles, in studies conducted by the Central Pollution Control Board (CPCB).

Water pollution has a direct impact on the quality and safety of our food supply. It is essential to address water pollution to ensure the health and well-being of both humans and the environment.



Solution to Water Pollution

We are all experiencing the impacts of water pollution, and it is essential to contemplate measures to reduce its increasing effects. To address water pollution, we need to start at the individual and household level. Minimizing the use of plastic is crucial, and carrying cloth or fabric bags when going to the market can significantly reduce plastic waste. Proper disposal of plastic and garbage is essential, avoiding throwing them into drains or on the streets.

Government support is vital in implementing effective measures. Proper treatment of liquid domestic waste and solid waste from cities should be ensured. Untreated waste must not be dumped into drains or rivers. Industries should be strictly prohibited from discharging untreated waste and encouraged to implement water reuse practices. Farmers also play a significant role in reducing water pollution. They should aim to minimize the use of chemical fertilizers and pesticides in their crops, promoting chemical-free agriculture.

Rainwater harvesting techniques should be widely adopted to prevent pollution from reaching groundwater. When clean rainwater replenishes groundwater, it helps in reducing contamination. Collecting various types of plastic in bottles and disposing of them in eco-bins can prevent plastic from reaching rivers. Additionally, the use of eco-bricks in parks and other areas can be an effective method. Schools can also implement the eco-brick approach.

By implementing these measures, we can collectively contribute to reducing water pollution and protecting our environment. It is essential for everyone to take responsibility and actively work towards a cleaner and healthier future.

What should I do ?

Water pollution is indeed a serious issue that requires our collective effort to find solutions. Each of us can contribute by practicing small precautions to partially prevent water pollution. Here are some points to keep in mind:

1. Do not litter or spread garbage in your surroundings.
2. Don't defecate in public spaces.
3. Consider installing a water purification system in your home or storing water in a copper vessel for three to four hours before consumption.
4. If the water from your hand pump or tap is contaminated, conduct tests to check for any contamination.
5. If the water from a hand pump is contaminated, try installing it at a deeper level. If it still provides polluted water, discontinue its use.
6. Boil and strain polluted water before using it.
7. Raise awareness among your friends and others about the impact of water pollution on health. If your village, town, or city is affected by water pollution, report the contamination to the relevant authorities.

By following these guidelines, you can contribute to reducing water pollution and protecting the environment. Remember, even small steps taken by individuals can make a significant difference in addressing this issue.

River

CHAPTER FOUR

A river is a flowing stream of water on the Earth's surface, typically sourced from a lake, glacier, waterfall, or rainwater, and it flows into an ocean or lake. The word "river" is derived from the Sanskrit word "nadī," and in Sanskrit, rivers are called "saritā." Rivers can be perennial or seasonal. Perennial rivers have their source in lakes, waterfalls, or glaciers and remain filled with water throughout the year, while seasonal rivers depend on rainfall. The Ganges, Yamuna, Kaveri, Brahmaputra, Amazon, and Nile are examples of perennial rivers, whereas the Kali, Hindon, Narmada, Godavari, and Krishna are examples of seasonal rivers. Rivers have always been the cradle of civilizations and have nurtured and supported human societies throughout history.

River System of India

Rivers in India have had economic, social, and cultural significance, which is still prevalent today. The development of the oldest civilizations took place in the valleys of the Ganges and Indus rivers. Even today, these two river valleys have the highest population and agricultural areas. Most of the major religious sites in the country are located along the banks of rivers. In ancient times, due to the convenience of trade and transportation, most of the cities in the country were situated along the rivers. India is often referred to as the land of rivers due to the abundance of rivers. There are mainly four river systems, or drainage basins, in India. In northern India, there is the Indus River system, the Ganges River system in central India, the Brahmaputra River system in northeastern India, and a network of various rivers in the peninsular part of India.

Broadly speaking, the river system of India can be divided into two categories

- Rivers originating from the Himalayas
- Rivers of the peninsular region

The rivers of the peninsular region can be further divided into three different categories

- Rivers originating from the south
- Coastal rivers
- Rivers originating from the intermountain regions

The river system of India is extensive and diverse, with evidence of ancient civilizations found along their banks, which still holds significance today. The importance of rivers is evident in the fact that major cities and highly populated areas are located along their banks. Rivers are considered life-giving because they are dynamic systems where water continuously flows. The flow of water in rivers helps maintain a highwater table and ensures the prosperity of the surrounding regions. Rivers not only provide drinking water but also serve agricultural purposes, supplying water for irrigation. Furthermore, rivers play a crucial role in the economy, supporting tourism and water transport.

In India, rivers hold not only economic significance but also religious importance. Many sages and saints have attained enlightenment sitting on the banks of rivers, and great personalities have contributed to society through the wisdom gained from these riverside experiences. The banks of rivers such as the Ganges, Shipra, and Godavari in India host the Kumbh and Ardh Kumbh fairs, attracting millions of people seeking spiritual and intellectual enlightenment. Rivers are an integral part of India's identity, and the concept of India is inseparable from its rivers. The nature of India and its rivers are deeply intertwined.

The rivers in India can be broadly categorized into two groups: Himalayan rivers and peninsular rivers. Himalayan rivers, such as the Brahmaputra and the Indus, originate from the Himalayas, extending beyond India's borders, and flow through Pakistan and Bangladesh, eventually merging into the Arabian Sea and the Bay of Bengal. Rivers like the Ganges and Satluj originate from the Himalayas within Indian territory and merge into the Arabian Sea and the Bay of Bengal after traversing various states in India.



There are also tributary rivers of both the Himalayan and peninsular rivers. These rivers originate in India and eventually merge with other rivers or flow into the ocean within India. However, the river Musi and its tributaries, originating in India, do not meet any other river or flow into the ocean. Instead, they are absorbed in the Rann of Kutch.

There are three major river systems associated with India in the Himalayas: the Indus River system, the Ganga River system, and the Brahmaputra River system. These rivers have numerous tributaries originating from the Himalayas, as well as from lakes, ponds, forests, and plateaus. Among the three, the Ganga River system holds the utmost importance, with approximately 2,071 kilometers of its total length of 2,525 kilometers flowing through Indian territory. The source of the Ganga River is also situated within the borders of India in the Himalayan mountains. The Ganga River predominantly flows through the plains, making it the lifeline of India, accounting for about 90 percent of its flow. The significance of the Ganga River is reflected in the fact that several rivers in different regions of the country are referred to as the Ganga, such as the "Ganga of Malwa" and the "Ganga of South India." After flowing through India, the water of the Ganga River nourishes Bangladesh and eventually forms the Sundarbans Delta in the Bay of Bengal. In Bangladesh, the Ganga River is known as the Padma River, and upon merging with the Brahmaputra River, it is referred to as the Meghna River. Hence, the river that forms the Sundarbans Delta in the Bay of Bengal is known as the Meghna River in Bangladesh.

The Brahmaputra River basin has a significant portion located outside India, originating in Tibet. However, the river holds greater importance in the northeastern states of India as they heavily rely on its waters for irrigation and hydroelectric power generation.

The Indus River system is formed by the convergence of three countries: China, India, and Pakistan. It irrigates the Indian regions of Jammu and Kashmir, Ladakh, Himachal Pradesh, and Punjab. The five major rivers of this system in India, namely Jhelum, Chenab, Ravi, Beas, and Sutlej, join together to form the Panjnad River in Bahawalpur, Punjab, which then merges with the Indus River 45 kilometers downstream. India benefits from abundant water and electricity through the entire Indus River system. In 1960, India and Pakistan signed the Indus Waters Treaty, which grants India the right to use the waters of the three rivers—Beas, Ravi, and Sutlej—for irrigation, power generation, and transportation purposes, while Pakistan has similar rights over the other three rivers—Indus, Jhelum, and Chenab. All the rivers in these three river systems of the Himalayan region contribute to their catchment areas.

A subcontinent refers to a land area surrounded by water on three sides, and the lower regions of India, particularly the central and southern parts, exhibit such geographical features. The mouths of some rivers in the subcontinent open into the Arabian Sea, while others flow into the Bay of Bengal. Most of the rivers originating from the Vindhyachal and Satpura mountain ranges, as well as the Amarkantak and Brahmagiri hills, merge with the Bay of Bengal. In between, hundreds of smaller rivers join these major rivers and lose their identity. Except for some rivers in Madhya Pradesh, all the rivers from Chhattisgarh to Andhra Pradesh, Karnataka, Telangana, Maharashtra, Tamil Nadu, Odisha, and Kerala fall under the category of peninsular rivers.

These rivers flowing through the Indian subcontinent strengthen the country economically, socially, and culturally. They provide India with drinking water, irrigation facilities, and important resources such as electricity and transportation. They also play a crucial role in addressing issues related to groundwater levels, water quality improvement, and solving problems of floods and droughts. A significant portion of India's economy depends on the uninterrupted and clean flow of these rivers. Currently, apart from the rivers in central India, most of the rivers have become seasonal. Although the water flow in the Himalayan rivers has decreased by about 40% due to factors such as reduced rainfall, they still maintain a

year-round water supply. Many of the assisting rivers in the Himalayan region have either disappeared, dried up, or turned into polluted drains. The conditions of the peninsular rivers have deteriorated even further. If a country's identity is intertwined with its rivers, and when rivers like the Ganga, Shipra, and Narmada flow, which hold mythological significance, deteriorate, human life and the environment, including biodiversity, will undoubtedly be in crisis.

What is “Apavah Tantra”?

The term “Apavah Tantra” refers to the system of flow or hydrology, specifically pertaining to the subterranean flow of water through streams and rivers. Apavah Tantra, or flow system, represents the specialized arrangement of water flow created by a river and its tributaries. It forms a network or framework where rivers converge to establish a unidirectional flow of water. All the contributing tributaries and other characteristics of a river basin together constitute its flow system.

From a scientific perspective, rivers perform three primary functions:

Drainage: The drainage function of a river involves processes such as infiltration, fluid movement, erosion, and sedimentation, which occur through the combined efforts of compression, friction, and collision.

River transportation: River transportation refers to the movement of substances being transported within a river, along with the flowing water.

River deposition: River deposition occurs when there is a decrease in the velocity of transported substances, leading to their subsequent deposition within the river.

Synonyms for Nadi (River)

English translation of “नदी” is “River”. Other synonyms are:

Payasvini, Tarni, Srotasvini, Sarita, Tatini, Lahari, Sari, Tarini, Tarangavati, Apaga, Nimga, Tarangini, Pravahini, Dvipavati, Laramala, Nadiya, Nirjharni, Jalamala, Shaivalini, Koolanksha, Nada, Sarita, Tatia, Kallolini, Sarang, Tanuja, Dariya, Tati, Kagolini, Dukoolani, Dukoolavati, Dharavati, Nirjhari, Shailaja, Srotasvini, Koolanksha, Kalkalnanadini, Samudraga, Stravanti, and Vahini

Delta

The area formed in a triangular shape by the accumulation of soil and sediment brought by rivers before merging into the sea is called a delta.

Peninsula

A peninsula is a region surrounded by the sea on three sides and connected to the land on one side.

Estuary

An estuary refers to a partially enclosed coastal body of water where freshwater from rivers mixes with salty seawater. It is the area where a river's fresh water meets the ocean's saltwater.

Gulf

A gulf refers to a part of the sea that is surrounded by land on three sides and connected to water on one side. Gulfs can be of two types: narrow and wide. A narrow gulf, which is longer in length, is called a "strait" in English, while a wider gulf is called a "bay."

Rift Valley

A rift valley is an area between two mountains that has sunk due to tectonic forces. The sunken part is called a rift or Drona Valley.

Water Reservoir

A water reservoir refers to a place where melted snow or rainfall water gathers through rivers, canals, and streams. Water reservoirs can be open or closed. An open water reservoir is where the water flows into the sea or ocean, while a closed water reservoir is where the water flows into lakes, marshy areas, or arid regions. Some closed reservoirs may become dry due to increased salinity over time, such as the Sambhar Lake in India. Additionally, some closed reservoirs may evaporate over time, leaving behind salt flats. Approximately 18% of the Earth's land falls under the category of closed water reservoirs. In India, the Ganga water reservoir is the largest, with nearly 20 rivers contributing to it.

Categories of Water Dividers

There are three categories of water dividers: continental, major, and minor.

Continental Water Dividers: Continental water dividers refer to those where the water from the same water catchment area flows in different directions. For example, the water of the Nile River in the African continent flows into the Mediterranean Sea, while the water of the Congo River flows into the Atlantic Ocean.

Major Water Dividers: Major water dividers are those where the water from different water catchment areas flows and converges at the same location. For instance, the water of the Huang He (Yellow River) and Yangtze rivers in China converge and meet in the Pacific Ocean.

Minor Water Dividers: Minor water dividers are characterized by the merging of water from different rivers at a particular location. For example, in India, the waters of the Ganga and Yamuna rivers merge at Prayagraj (Allahabad).

A plateau is a specific location on the Earth's surface that is sufficiently elevated from the surrounding land. Its upper part is broad and flat, and this region is known as the plateau. Plateaus are usually around 600 meters higher than sea level. In other words, we can say that the flat land on top of mountains is called a plateau. The classification of plateaus is not solely based on elevation. Plateaus are of great importance as they form secondary category habitats on land. Approximately 33 percent of the Earth's surface consists of plateaus. The Pamir Plateau is the highest plateau in the world, also known as the "Roof of the World."

Some major reasons for the formation of plateaus are as follows:

- Tectonic movements that cause a flat land area to rise above the adjacent surface.
- Upheavals that result in the subsidence of neighboring land areas while keeping a flat portion elevated.

- Formation due to the solidification of lava emitted during volcanic activity.
- Formation due to the inability of nearby areas to rise higher during mountain-building processes.

Plain

Plains From a geographical perspective, a plain refers to a land area that is either flat or has slight undulations and is at an elevation of approximately 500 feet or less from sea level.

Doab (interfluve)

A doab refers to the region between two rivers. It is formed by the combination of the words “do” meaning two and “aab” meaning water, indicating a land between two water streams, such as the land between the Ganges and Yamuna rivers. There are several such regions in the world. The Doab specifically refers to the land between the Ganges and Yamuna rivers, which stretches from the Shivalik Hills in Uttar Pradesh to the confluence of both rivers in Prayagraj.

Doab regions in the Indian subcontinent:

- Ganga-Yamuna Doab
- Indus-Jhelum Doab
- Jhelum-Chenab Doab
- Chenab-Ravi Doab
- Ravi-Beas Doab
- Beas-Sutlej Doab



How to Conserve Water

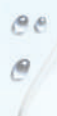
CHAPTER FIVE



To manage the tasks related to water conservation within the region, we need to consider each district as a unit within the province. Therefore, to carry out the tasks of water management, we need to implement a program called “My District-My Initiative.” Under this program, work will be done for water conservation and river protection. In each district, in collaboration with the community, various activities that contribute to water management will be undertaken. These activities will include local initiatives, public awareness, and policy-based actions. Some of the tasks will yield long-term results, while others will show immediate effects.

My District-My Initiative

The “My District-My Initiative” is initiated as a social endeavor with the aim of engaging society in water conservation and river revitalization efforts in their respective districts (villages and towns), in conjunction with the government’s positive initiatives. All the tasks carried out under this program will be done with the cooperation of the community and local administration. This social initiative will allow every individual to contribute in any way possible to the restoration of ponds, wells, and rivers.



My District-My Initiative Framework

A central committee for “My District - My Initiative” will be formed, consisting of five members. Additionally, each village, town, or city in the district will have its own committee with five members. The district committee will be formed by the district team, while the village, town, or city committees will be formed by the district committee with the approval of the regional committee. All committee members will be individuals who possess a social

conscience, work in this field, are subject matter experts, and retired officials, all of whom will serve voluntarily.

The district committee will consist of five members, including one committee chairperson and four other members. Among the four members, one will be responsible for public awareness, two for ground-level activities, and one for documentation and media.

Village/Town/City Committee: Each committee will consist of one committee chairperson and four members. Among the four members, one will be responsible for public awareness, two for ground-level activities, and one for documentation and media.

Volunteers of My District - My Initiative

A central committee for "My District - My Initiative" will be formed, consisting of five members. Under this program, volunteers from "My District - My Initiative" will be appointed in all villages, towns, and cities of the district. Any volunteer can be an ordinary person who believes in contributing to water conservation and river rejuvenation in their respective village, town, or city. These volunteers can contribute from anywhere and in any way they can. They can either work according to the district's plan or inform the district committee about their own plan. These volunteers will serve as ambassadors for water conservation and river rejuvenation in the district. Regular volunteer conferences will be organized at their convenience, and an annual conference will be mandatory.

- Volunteers can perform the following tasks with the support of the Village/Town/City and District Committees:
- Conduct cleanliness programs in rivers in villages, towns, and cities.
- Support the revival of ponds in villages, towns, and cities.
- Identify areas in villages, towns, and cities affected by water issues such as water pollution and scarcity, and work towards their resolution.
- Collect data related to water sources in villages, towns, and cities.
- Organize water awareness sessions in society to promote the judicious use of water in villages, towns, and cities.
- Conduct surveys of families in villages, towns, and cities facing serious water pollution problems or experiencing severe illnesses, and make efforts to provide clean drinking water through local administration, government, or other organizations.
- Organize awareness programs in private and government schools in villages, towns, and districts.

Note: It has been observed that both state and central governments have implemented various schemes at their respective levels for rivers, ponds, water, agriculture, and other related issues. Despite the presence of departments at the district, tehsil, and block levels, the water-related infrastructure continues to deteriorate. There are four main reasons behind this: firstly, inadequate implementation of schemes at the ground level; secondly, lack of awareness among local communities about the schemes; thirdly, ineffective performance of government departments; and fourthly, widespread corruption. In such a

situation, the role of volunteers becomes crucial. In order to work towards the rejuvenation of rivers and ponds, it is essential to ensure the proper implementation of all government schemes on the ground, adhering to the relevant rules and regulations.

Water Conservation Work

As per the decision, in order to address the water crisis, we need to approach the problem through three strategies: public awareness, policy measures, and on-ground efforts. To begin this work, it is crucial for us to gather comprehensive information from all perspectives regarding the tasks and related aspects. Therefore, to carry out water-related projects in our region, we must have complete information about all the relevant issues. For effective implementation of water initiatives in each district, it is necessary to have data on natural water sources (such as ponds, rivers, and wells) as well as human-made sources (such as canals and tube wells) present in all the districts of our region. Understanding the specific problems associated with each location is also essential. This includes information on groundwater and surface water availability, annual rainfall, water pollution status, agricultural practices, and other relevant water-related statistics. In order to transform our region into a water-rich area, it is imperative to approach the work at the district level, considering each district as a unit. Therefore, any plans or actions implemented will be carried out at the district level. To rejuvenate ponds and wells in each district of the region, a large-scale program must be implemented, requiring the collaboration of both society and the government. To gather information about the ponds and wells in your district, the following questions need to be addressed.

- How many ponds are registered according to the revenue records in the district?
- What is the total area of ponds recorded in the revenue records of the district?
- What are the survey numbers of the existing ponds according to the revenue records?
- How many ponds are currently present?
- How many of the existing ponds have water?
- How many ponds are located within the populated areas, and how many are outside the populated areas, i.e., in the forest?
- How many ponds are under encroachment (partially or fully), and what type of encroachment is it (permanent or temporary)?
- How many ponds are contaminated?
- How many ponds are used for commercial activities such as fish farming and water chestnut cultivation?
- How many wells are there in the district?
- What is the current condition of the wells?
- How many wells are within the populated areas, and how many are outside in the forest?

Some answers to the above questions can be obtained from the district administration, but for the remaining unanswered questions, a survey will need to be conducted in each village.

The need for pond rejuvenation

In the country, there are approximately 40 lakh ponds and water conservation structures related to around 6,64,369 villages, towns, and cities in about 718 districts. By rejuvenating these ponds, we can alleviate nearly 50 percent of the water crisis in the country.

According to the NITI Aayog (National Institution for Transforming India), in the past decade, hundreds of rivers and millions of ponds and wells have dried up in our country. If the situation continues as it is, by the year 2030, nearly 40 percent of the population will not have access to potable water. Extensive research has established that ponds play a vital role in maintaining a continuous flow in rivers and sustaining wells. However, in recent decades, we have almost forgotten the significance of ponds, and we have completely eliminated about 30 percent of them. The remaining 70 percent of ponds are either in poor condition or contaminated with polluted water, leading to their deterioration and the spread of diseases in villages. If we assess the severity of the situation, we have nearly destroyed natural water sources. The ponds and lakes that used to preserve clean rainwater and recharge groundwater now accumulate highly polluted water due to population growth, resulting in groundwater contamination.

In recent times, our governments have shown a keen interest in rejuvenating ponds, which is a positive sign. However, the methods being employed for pond rejuvenation are not adequate because the issues concerning ponds have evolved. The current approach often involves deepening the ponds with the aim of rejuvenation, but this leads to the accumulation of even more polluted water due to the increased population, further contaminating the shallow groundwater."

Now the question arises: What is the correct method for pond rejuvenation? Our country has several educational and research institutions that have made significant efforts and carried out various groundlevel initiatives for pond rejuvenation. Among the efforts made, one successful method is the approach adopted by IIT Mumbai, which has also been implemented by NIE Roorkee. Another method has been developed by NEERI (National Environmental Engineering Research Institute).



How to undertake lake rejuvenation work?

The following precautions or information are necessary for lake rejuvenation:

Firstly, gather the official documents (lake ownership papers and any maps) of the selected lake for rejuvenation. This will provide you with information about the lake's area and its plot number.

The selection of the lake should be done with the consent of the local governing body, such as the village council, municipal council, or city corporation, and the representatives and officials of the administration.

- Assess the current condition of the lake by visiting the site.
- Is the lake situated in a populated or non-populated area?
- What is the significance of the lake, its name, and its history?
- Is the lake encroached upon? If yes, to what extent?
- If the lake is encroached upon, how much of it is temporary encroachment and how much is permanent?
- Is there any legal dispute regarding the lake in a court of law?
- Is the lake polluted or contaminated?
- Is the lake dry or filled with water?
- Are there any aquatic plants, grass, or water guardians in the lake or not?
- How many routes are there for water to enter the pond?
- What is the quantity of water entering the pond out of 100 ponds?
- What is the quality of the water entering the pond?
- What other things come along with water through channels in the pond, such as cow dung and plastic, etc.?

After obtaining all the above information about the pond, a proper strategy can be determined to revive the pond. How much will it cost, and how long will it take to complete the work? If there is any legal dispute regarding any pond, one should refrain from working on it. Before carrying out the pond revival work, photographs of all four corners of the pond should be taken so that a visual representation of the transformation can be presented to society once the pond revival work is completed. This boosts the morale of the workers and the organization involved in the work, and it also inspires others through good work.

If the pond is filled with water and sludge

In such a situation, first completely drain the pond using a pump or a motor and prevent any water from entering the pond through all the channels until the sludge in the pond is completely dry. For this, a pit can be dug on one side of the pond to hold the water and prevent it from entering the pond until the sludge dries up in other parts of the pond. Once the work in the other parts of the pond is complete, improvement work in the pit area can be carried out.

If the pond is filled with water hyacinth, grass, and other floating debris

It needs to be determined whether the water in the pond has become stagnant or the pond has dried up due to water hyacinths, grass, or other floating debris. If so, the water should be pumped out using a pump or a motor. The water hyacinths and grass removed from the pond can be used as good organic fertilizer. For this, a liquid bioenzyme spray can be applied to them. Local agriculture departments can also be contacted regarding this.

In both of the above situations, after removing the water from the pond, it should be left to dry for one week. Within one week, the sludge in the pond will reach a state where it can be removed by machines or laborers. The sludge should be removed until the sandy soil is exposed in the pond. The upper layer of sludge coming out of the pond should be disposed of away from the pond, while the lower, moist, and dry sludge or soil should be used to create the pond bund. The height of the bund should not exceed ten feet, although it can be adjusted according to the circumstances and can be up to twenty feet. The inner part of the bund should have a gentle slope. Ideally, the outer part of the bund should also have a gentle slope, but if that is not possible, it should not cause any problems. The bund should be curved along the edges of the pond. Proper arrangements, such as pipes or concrete drains, should be made for the water to enter the pond from the routes where water flows into the pond. A 15 to 20 feet wide pathway should be left on one side of the pond for cleaning, commercial activities, and easy movement of animals.

If the pond is completely dry

Reviving a completely dry pond is easier compared to filling a pond with water. In this case, two to three feet of soil should be removed from the pond's bed. The excavated soil can be used to create the pond bund. Working on such a pond is easier with laborers.

The method of pond rejuvenation

To implement the method of pond rejuvenation, the channels that bring water to the pond should be connected and directed to a single location. For this purpose, a canal can be



constructed on the outer edge of the pond. According to the given information, appropriate arrangements should be made to treat the various types of impurities that enter the pond along with the water, so that no type of dirt enters the pond apart from water. It would be better to adopt the three-pit purification technique for this task.

According to this method, the pond is divided into two parts. The first part consists of approximately 15% of the total pond area and is developed for the purification of dirty water using a natural water purification system. The remaining portion is used for constructing the pond, which continuously collects the purified water from the first part. The water collected in the pond is used for various purposes such as fish farming, irrigation, and groundwater recharge. The pond is organized in such a way that it can collect rainwater and store the incoming water regularly in a proper manner. This pond rejuvenation system not only revitalizes the pond but also serves as a regular means of employment, such as fish farming and the production of water chestnuts (singhada), which can be carried out in the pond.

Plantation on the bunds

Plants should be planted between the bunds constructed along the edges of the pond. It is best to plant trees such as neem, jamun, sheesham, peepal, banyan, pilkhan, and guava. A distance of at least ten to twenty feet should be maintained between each plant.

Provision of water in the pond

The ponds located within the settlement area receive water from the village itself. The ponds located outside the settlement area used to be filled with rainwater or mostly with water from canals. In most ponds, small channels are documented in government records to bring water from the canal to the pond. The village patwari (revenue officer) used to record the time it took to fill each pond from the canal. When working on such ponds, efforts should be made to connect them to the nearby canal through a channel so that the excess rainwater can also enter the pond.

Ponds should not be filled with groundwater

The purpose of a pond is to collect not only village water but also rainwater. Therefore, the function of a pond is to gradually replenish the groundwater by slowly releasing the collected water. This water is used for various village activities such as bathing, providing water to animals, raising seedlings, fish farming, washing clothes by dhobis (laundry workers), and watering crops. The most important task and benefit of a pond is to naturally purify the water it contains and replenish the groundwater over time.

In this regard, filling the pond by extracting groundwater through tube wells or similar methods goes against the essence of the pond. Such activity cannot be justified in any way. Permission for activities such as fish farming or seedling cultivation should also be granted only in ponds where surface water can be replenished through channels or similar means. Ponds should not be filled with groundwater for any commercial activities, and the government should impose restrictions on this as well.

Put up a board at the pond

A board providing complete information about the pond should be installed at its shore. The board should contain the following information:

- Name of the pond.
- History of the pond.
- Area of the pond.
- Khasra number (land parcel number) of the pond.
- Water storage capacity of the pond.
- Estimated amount of water entering the pond in one year and estimated amount of water discharged from the pond in one year.
- Date of pond rejuvenation.
- Cooperation in pond rejuvenation.

Note: The work of pond rejuvenation should be done either before the rainy season or after the end of the rainy season

How to rejuvenate a river?

To rejuvenate a river, we first need to identify the rivers in our district. It is likely that a river originates from the same district and has its endpoint within the district. In most districts, we may observe that a river flowing through any part of the district originates from another district and continues into other districts after covering some distance within our district. It is also possible that a river starts and ends within the same district. Keeping these four scenarios in mind, we need to focus on river rejuvenation efforts. For this, we need to gather the following information:

- Which rivers originate from the district, pass through the district, reach the district, or flow entirely within the district?
- What is the history of these rivers?
- Do we have any maps of the river(s)?
- What is the length of the river?
- How many villages/towns/cities are situated along its banks?
- What are the current issues faced by the river, such as drying up or pollution?
- What are the causes of pollution in rivers?
- What actions can we take at our level to improve the river and establish its complete course?
- To carry out river rejuvenation work, it is essential to develop both immediate and long-term effective plans and implement them in an organized manner.
- River rejuvenation is not a task that can be accomplished in a few days, so we need to continue working on it. Both government and societal support are necessary for this.

endeavor. A model for river rejuvenation has been developed, and we should implement this model for the rejuvenation of each river, which includes the following steps:

River Rejuvenation Model

The existence of small and seasonal rivers in India is almost nearing its end. The new generation now considers these small rivers as domestic and industrial drains. On one hand, where the rivers that used to flow with water throughout the year are now experiencing water scarcity, the seasonal rivers have dried up due to the declining groundwater level. In this situation, where the survival of rivers is at stake, their identity has also changed. A notion has developed among the older generation that seasonal rivers cannot be rejuvenated, but that is not true. To transform the disappointment prevailing in society regarding small and seasonal rivers into hope, this model has been developed. The results of successful efforts made on the primary tributaries of the Ganges, such as the Eastern Kali and Neem rivers, and the tributary of the Yamuna, Hiran, have given us the energy, enthusiasm, and confidence that India's seasonal rivers can be revived. However, it requires systematic efforts. River rejuvenation initiatives are being undertaken in various parts of the country. Based on the work done and the expertise gained over the past fifteen years, this organized model for river rejuvenation has been created. This model can be applied to rejuvenate any small river in the country.

River Rejuvenation Model

It is a very simple model, and some results from the suggested measures start showing immediately, while others take a long time. This model not only brings about the rejuvenation of the river but also enhances the well-being of the communities living along the riverbanks. In order to implement this model, activities River Rejuvenation Model within a one-kilometer radius from both sides of the river, where the river is a significant part, are also highly impactful. The suggested actions in this model can only be successful through collective efforts and cooperation between society and the government. The River Rejuvenation Model has defined 10 fundamental principles:

1. Knowledge of the river.
2. Identification of riverine land.
3. Revival of river sources.
4. Management of liquid and solid waste.
5. Cleaning of river channels.
6. Public awareness.
7. Conservation of ponds.
8. Dense afforestation.
9. Chemical-free agriculture.
10. Construction of small dams.

Knowledge of the river

Understand the river before taking action.

Only then can you effectively contribute to its well-being

The following information about the river is necessary for initiating the task of its improvement or rejuvenation:

- History of the river (folklore and scientific perspectives).
- Social and religious significance of the river.
- All documents related to the river (governmental/non-governmental).
- Understanding the origin of the river.
- Knowledge of the length and catchment area of the river.
- Water conditions of the river.
- Current issues faced by the river.

It is particularly important to understand the river's origin correctly. The flow of every monsoon river is a result of the confluence of groundwater and surface water. Surface water keeps meeting groundwater at the location where the groundwater level is at its highest (signifying the occurrence of water bubbles on the ground surface). Therefore, this area is the lowest and the one with the highest water levels. The regions where there has been a continuous presence of abundant water flow are developed as the source areas of monsoon rivers. Over the past 4 to 5 decades, the source areas of monsoon rivers have remained water-rich and free from encroachment. As the groundwater level gradually declined, the local residents encroached upon vacant land, leading to the gradual encroachment of the source areas of monsoon rivers.

Identification of River Land

When a river is free from encroachments.

Only then will it become water-bearing.

Most of the land along the banks of monsoon rivers is somehow associated with some form of water body. Wherever a monsoon river originates, there will be a pond, lake, reservoir, vacant land, or agricultural activities present. Even in the total length of the river's flow from its origin, both banks of the river will be encroached upon. In some places, farmers have made their fields by merging them with the river, while in others, someone else has encroached upon river land. In some areas where the catchment area of the river is larger, farmers start engaging in agricultural activities within the river as well. In such cases, it is necessary to free the land on both sides of the river, just as the river's origin is kept free. For this, coordination should be established with the local administration, and the river's origin and catchment area should be identified by obtaining accurate information about the river.

Which districts does the total area covered by the river's length pass through? Accordingly, this work will have to be carried out through the administration of different districts. The measurement of river land should be done on both banks of the river. If possible, pillars or markers should be installed on the marked river land.

River rejuvenation at the source

*When there is water at the source,
Only then will a river flow.*

The birthplace of any river is marked by the presence of water, so it is essential for the source of the river to have water. Therefore, efforts to rejuvenate the river should be initiated at the source. This task can be carried out in the following manner:

- Identification of the source of the river based on revenue records, gazetteers, and other related documents.
- Inspection of the source of the river.
- Verification of the extent of land registered in revenue records as part of the river's source. Is the same amount of land present at the site? This information may not be available to us, so it would be necessary to measure the land at the source of the river with the assistance of the concerned sub-district officer/tahsildar. The cooperation of the higher authorities, dignitaries, and representatives of the area may also be sought for this task.
- If there is no encroachment on the land at the source of the river, it will be evident through the team from the sub-district.
- If there is any encroachment on the land at the source of the river, the first task would be to reclaim that encroached land. There should be no domestic sewage or untreated industrial waste flowing into the river's source. If this is happening, the assistance of the local pollution control department would be required to control or resolve the pollution.
- If there is neither encroachment nor pollution at the source of the river, then work should be initiated immediately. Let's assume that the river's source has 10 hectares of land; in that case, an artificial lake with a depth of 8 to 10 meters should be constructed, leaving a pathway of ten meters around it. For water to flow into the lake during rainfall, a pathway for water to enter from beneath the embankment is necessary. It will be determined on the land at the source of the river how lakes, ponds, and dams can be constructed. For this, we can seek advice from the local irrigation department, irrigation department, or any relevant experts.

The source of any monsoon river should be such that it can collect as much rainfall as possible. If it is possible to divert additional water from canals or distributaries, that can also be done. Adequate arrangements should be made.

Liquid and Solid Waste Management

*When there is no garbage overflow in the river,
then the river will become clean and flowing.*



Liquid and solid waste management is not limited to social organizations or other groups working for the river, as preventing domestic and industrial discharge is a complex process that only the government can handle. This work involves numerous rules and regulations and carries a high financial burden.

In such cases, social organizations or groups can assist by implementing natural water purification methods to prevent the flow of polluted water from villages into the river.

However, while villages may have a smaller role compared to towns and cities, awareness work should still be carried out in the villages. Strict regulations should be imposed on industries that dispose of their liquid and solid waste into the river, and the government needs to take concrete steps to enforce these regulations. Each industry should have an Effluent Treatment Plant (ETP) installed and properly operated. Additionally, towns and cities should have Sewage Treatment Plants (STPs) for liquid waste and proper arrangements for solid waste disposal. Solid waste management systems should also be implemented in each village and city.

River Flow Cleaning

*When the course of the river is clean,
only then will the water flow uninterrupted.*

Cleaning the main course of the river in its flow area is essential to maintain the flow of any river. It is often observed that the flow areas of small rivers remain clogged with dirt or are illegally encroached upon. In such cases, it is necessary to clean the river flow area from

both banks and the riverbed. The responsibility for cleaning the flow area of the river that falls within the boundary of a village/town/city can be assigned to the respective entity to ensure better execution of this task. The dirt flowing out of the river should not be dumped on its banks but should be deposited far away from the river to prevent its reentry through any means. Clearing the path of the river's flow is also crucial for maintaining continuous and clean water flow. This task can be accomplished by involving the villages along the riverbank under the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA).

Public Awareness

*When hands reach out from village to village,
the work of river restoration will flourish.*

Efforts to raise public awareness about rivers should be promoted in villages, towns, and cities located within a kilometer's distance from both banks of the river. This can be achieved through street plays, wall writings, competitions in schools, inter-village visits, street meetings, and public awareness gatherings. Information about all the activities that an ordinary citizen can undertake for the rejuvenation of their river should be disseminated through public awareness campaigns to reach the villages, towns, and cities. A program of public awareness should be implemented in such a way that all residents along the river's banks willingly fulfill their responsibilities towards the river. Public awareness should cover all the topics necessary for river restoration among the villagers, including organic farming, afforestation, pond conservation, sustainable water usage, waste management, and cleanliness.

Dense Afforestation

*Let the riverbank not remain empty;
Trees shall be its guardians.*

Within a one-kilometer radius from both banks of the river, dense afforestation should be carried out on the river's land and floodplains. The plantations should consist of species that can retain water in their roots and prevent soil erosion. Special programs should be implemented in villages within this radius to encourage farmers to establish orchards and plant trees on their agricultural land. By doing so, not only will there be increased rainfall in the area within a few years, but the groundwater level will also rise, assisting the river in maintaining a steady and clean flow. The local forest department, community, and social organizations should actively collaborate in the efforts of dense afforestation.

Lake Conservation

*Lets pick up the tools;
make our lakes clean.*

In villages within a one-kilometer radius from both banks of the river, when the lakes are clean and filled with water, the rising groundwater level will gradually reduce the pollution present in the groundwater. The increased groundwater level will facilitate the river's flow, and all the lakes in the villages within a onekilometer radius from both banks of the river should be identified and rejuvenated based on revenue records. If there are any ponds or reservoirs along the riverbank, they should also be rejuvenated. If there is any government land available along the riverbank where a lake can be constructed, it should be done with the support of the local administration.

Chemical-Free Agriculture

*When crops are chemical-free,
the river will flow with purity.*

Agriculture within a one-kilometer radius from both banks of the river needs to be made chemical-free. The use of any chemical fertilizers and pesticides in any crop should be discontinued, and farming should be done using natural methods. This will result in crops that are pure and nutritious, and it will also improve the quality of the soil. Since the Green Revolution, the continuous use of chemicals has led to their slow infiltration into the groundwater. Chemical-free agriculture will improve the quality of groundwater and also prevent the elements of fertilizers and pesticides from flowing into the river through irrigation water or rainfall. The relevant agricultural schemes of the district can also benefit from this approach.

Construction of Small Dams

*When dams are built along the river,
only then will the rivers flow be filled.*

Once complete removal of liquid and solid waste from the river is achieved, small dams should be constructed at specific distances based on the circumstances. These dams should be built in areas where there is no pollution in the river. Through these dams, we will be able to retain water on the river's surface and successfully replenish the groundwater. This will provide water to local residents, birds, and animals. The construction of dams on the river should be done with technical assistance from local irrigation departments, land conservation departments, irrigation departments, or subject matter experts.

Note: If 80% of the work is completed based on the above 10 fundamental principles for a river, then the river will return to its existence within 5 to 10 years. It is necessary to prepare a detailed outline specifically for the river to implement all 10 fundamental principles.

References

To gather the content of this book, information has been collected from reliable sources, which are as follows:

1. Center for Science and Environment, New Delhi.
2. Ministry of Environment, Forest and Climate Change, Government of India.
3. Ministry of Jal Shakti, Government of India.
4. Wikipedia.
5. Neer Foundation.
6. Environmental Conservation Activities.
7. Shri Krishna Gopal Vyas, India Water Portal.



Raman Kant - Riverman of India



Raman was born on 5 November 1980 in the historical Poothi village of Meerut district. After receiving his primary education from the village school, Raman passed his graduation examination from Chaudhary Charan Singh University. Raman has also done one year computer diploma and one year journalism diploma. Raman became involved in river rejuvenation work in the year 2001 itself. While Raman is doing groundwork by connecting with the society, he is also contributing to the policy schemes of the Central and various state governments. Work was started with full dedication at both the policy and ground levels towards keeping the natural water sources (ponds and rivers) pollution free and reviving them. Due to his river rejuvenation work of about 22 years, Raman is today known as Riverman of India in the country.

Raman is doing all his work with the spirit of 'Vasudaiva Kutumbakam' with a positive spirit in collaboration with the society and the government. According to Raman, when we get many things from nature including water, energy, food, and air, then we should also pay its price by conserving nature through some means or the other. Every person of the society should understand about his duty towards nature and should realize the concept of development without harming nature. Raman is working towards inculcating nature protection among children, diverting farmers towards chemical-free agriculture and establishing the relationship between water and women among women. Natural Environmental Education and Research (NEER Foundation) was formed by Raman in the year 2004.

Raman has expertise in river rejuvenation and pond and rainwater conservation. Raman is also proficient in the art of pencil art.

Influenced by Raman's river rejuvenation work, the first Bollywood film 'Ek Ank' has been made giving the best message of river conservation. The director of this film is Prabhat Kumar. The producer of the film is Dr. Neeru Joshi. The hero in the film is Yajuvendra Pratap Singh and the heroine is Preeti Shukla. The film has three songs sung by Sonu Nigam, Papon and Palak Muchachal respectively, which have been released by Zee Music. The music of the film has been given by Puneet.

Raman considers Lord Shiva as nature. All dimensions of Lord Shiva are symbols of nature. On this basis, Raman is slowly moving forward with full dedication. Thousands of volunteers from hundreds of villages, towns and cities are associated with them. With the main aim to restore the lost identity of Ganga-Yamuna Doab, Raman is moving towards the goals with a steady pace, hard work, dedication, and honesty.

